

☐ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. Project No.: WW-4705R
Project Name: Hilo WWTP Phase 1 Submittal Number: 11312R-001.0
Submittal Title: SINGLE-LOBE ROTARY PUMPS
TO: County ☐ For Information Only
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec: 11312R	Paragraph: 1.04 1.05 1.06 2.01 2.02 2.03 2.04 2.05 2.06 2.07 2.08 2.09 2.10 2.11 2.1:
Authorized By: Hawaii Engineering Services	Date Submitted: 09/04/2025

Submittal Certification	
Check Either (A) or (B):	
☒ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: 	
Printed Name and Title: Makoa Ng, QC, Project Engineer	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/4/2025
SPECIFICATION SECTION # 11312R
SPECIFICATION SINGLE-LOBE ROTARY PU
PARAGRAPH 1.04 1.05 1.06 2.01 2.02 2.03
DRAWING na
SUBCONTRACTOR na
SUPPLIER Boerger
MANUFACTURER Hawaii Engineering Servic

CERTIFIED BY CQCM or Designee : 

SECTION 11312R
SINGLE-LOBE ROTARY PUMPS

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes: Rotary lobe type positive displacement pump with driver and features as specified and scheduled in this Section. Refer to related Specifications for additional requirements.
- B. Tag numbers: As specified in Pump Schedule.

1.02 REFERENCES

- A. American Gear Manufacturers Association (AGMA).
- B. American National Standards Institute (ANSI).
- C. ASTM International (ASTM):
 - 1. A48 - Standard Specification for Gray Iron Castings.
 - 2. A276 - Standard Specification for Stainless Steel Bars and Shapes.
 - 3. A278 - Standard Specification for Gray Iron Castings for Pressure-Containing Parts for Temperatures Up to 650°F (350°C).
 - 4. A283 - Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates.
 - 5. A470 - Standard Specification for Vacuum-Treated Carbon and Alloy Steel Forgings for Turbine Rotors and Shafts.
 - 6. A516 - Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate and Lower Temperature Service.
 - 7. D2240 - Standard Test Method for Rubber Property-Durometer Hardness.
- D. Hydraulic Institute (HI):
 - 1. 3.1-3.5 - Rotary Pumps.
 - 2. 3.6 - Rotary Pump Test.
 - 3. 9.1-9.5 - Pumps - General Guidelines.
- E. National Electrical Manufacturers Association (NEMA).

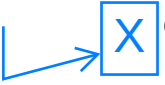
1.03 DEFINITIONS

- A. Pump head (total dynamic head, TDH), flow capacity, pump efficiency, net positive suction head available (NPSHa), and net positive suction head required (NPSHr) are as defined in HI 3.1-3.5, 3.6, and 9.1-9.5 and as modified in the Specifications.
- B. Suction head: Gauge pressure available at pump intake flange or bell in feet of fluid above atmospheric. Average when using multiple suction pressure taps, regardless of variation in individual taps.

1.04 SUBMITTALS

- ✓ A. Submit as specified in Section 01330 - Submittal Procedures.
- ✓ B. Vibration analysis: Submit as specified in Section 15050 - Common Work Results for Mechanical Equipment when scheduled.
- ✓ C. Product data: As specified in Section 15050 - Common Work Results for Mechanical Equipment.
- ✓ D. Shop drawings: As specified in Section 15050 - Common Work Results for Mechanical Equipment.
- ✓ E. Calculations: As specified in Section 15050 - Common Work Results for Mechanical Equipment:
 - 1. Torsional analysis: When scheduled, submit as specified in Section 15050 - Common Work Results for Mechanical Equipment.
- ✓ F. Vendor operation and maintenance manuals: As specified in Section 01782 - Operation and Maintenance Manuals.
- ✓ G. Commissioning submittals:
 - 1. Provide Manufacturer's Certificate of Source Testing as specified in Section 01756 - Commissioning.
 - 2. Provide Manufacturer's Certificate of Installation and Functionality Compliance as specified in Section 01756 - Commissioning.

1.05 QUALITY ASSURANCE

- ✓ A. As specified in Section 15050 - Common Work Results for Mechanical Equipment.
- ✓ B. Provide pumps specified in this Section from same manufacturer.
-  C. Vibration analysis (Rotodynamic Analysis): Provide equipment vibration analysis as specified in this Section:
 - 1. Vibration analysis level: As scheduled and as specified in Section 15050 - Common Work Results for Mechanical Equipment.
 - 2. Vibration analysis expert: Provide when scheduled.
- ✓ D. Vibration testing: Provide vibration testing as specified in this Section:
 - 1. Vibration testing level: As scheduled and as specified in Section 15958 – Mechanical Equipment Testing.
- ✓ E. Manufacturer's Certificate of Installation and Functionality Compliance as specified in Section 01756 - Commissioning.

1.06 WARRANTY

- ✓ A. As specified in Section 01783 - Warranties and Bonds.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- ✓ A. Pumps: One of the following:
 - 1. Boerger, LLC.
 - 2. Vogelsang USA, LTD.

2.02 DESIGN AND PERFORMANCE CRITERIA

- ✓ A. Components:
 - 1. Rotary lobe type positive displacement pumps with convoluted type dual or quad lobe rotors, motors, seals, couplings, base plates, guards, supports, anchor bolts, taps, lifting eyes, stands, and other items as specified and as required for a complete and operational system.
- ✓ B. Design requirements:
 - 1. Pump performance characteristics:
 - a. As specified in the Pump Schedule.
 - b. Performance tolerances shall be the same as the test tolerances specified in Section 15958 - Mechanical Equipment Testing.
 - c. Pumps shall be suitable for installation as specified in this Section and as indicated on the Drawings.
 - 2. Motor characteristics: As specified in Pump Schedule.
- ✓ C. Product requirements as specified in Section 01600 - Product Requirements and Section 15050 - Common Work Results for Mechanical Equipment.

2.03 MATERIALS

- ✓ A. General: Materials in the Pump Schedule shall be the type and grade as specified in this Section.
- ✓ B. Cast iron: ASTM A48, Class 30 minimum.
- ✓ C. Gray iron casting: ASTM A278, Class 30.
- ✓ D. Buna-N: Synthetic rubber with a minimum Durometer hardness of 70 in accordance with ASTM D2240 test methods.
- ✓ E. Steel: ASTM A283, Grade D or ASTM A516 Grade 70.
- ✓ F. Stainless steel: ASTM A276, Type 316 stainless steel; nickel - chrome - boron coating as scheduled.
- ✓ G. Carbon steel: ASTM A470.
- ✓ H. All elastomers shall be Buna-N.

2.04 GENERAL PUMP CONSTRUCTION

- ✓ A. Type: Industrial, heavy duty, positive displacement, rotary lobe type pumps meeting performance requirements and features as scheduled and as specified.
- ✓ B. Service: Pumping units shall be designed to convey blended sludge, which is a combination of primary sludge, trickling filter humus/waste activated sludge, and scum. Normal solids concentrations will range from 1 to 2 percent. Other service requirements shall be as scheduled.
- ✓ C. All equipment shall be designed and built for 24-hour continuous service at the rated design condition without overheating, without cavitation, and without excessive vibration or strain.
- ✓ D. All working parts of the pumps and motors, such as bearings, wearing rings, shaft, sleeves, etc., shall be standard dimensions built to limit gauges or formed templates, such that parts will be interchangeable between like units and such that the Owner may obtain replacement and repair parts for those furnished in the original machines at any time in the future.
- ✓ E. All lubrication fittings shall be brought to the outside of all equipment so that they are readily accessible from the outside without the necessity of removing covers, plates, housings, or guards.
- ✓ F. Pump shall be capable of temporarily running dry without damage and operate in either direction.

2.05 PUMP CASING

- ✓ A. Materials:
 - 1. Pump casing: As scheduled.
 - 2. Pump top and bottom housing: As scheduled. Minimum 650-725 Brinell hardness.
 - 3. Front and end covers: Same as pump casing.
 - 4. Wear plates: As specified.
- ✓ B. Top and bottom pump housing: Top and bottom segments of the pump shall be adjustable based on wear up to 6 millimeters. The adjustment shall be accomplished by simply moving stainless steel shims from one hole to the next in the pump housing, allowing for the closing of tolerance around the rotors. This adjustment must be available a minimum of 2 times from factory tolerance:
 - 1. In lieu of providing adjustable top and bottom segments, manufacturer may provide radial wear plates on the casing walls that utilize the same philosophy as the front and rear wear plates.
 - 2. The radial wear plates shall be designed to allow the Owner to replace the worn component and bring the radial casing back to factory tolerance.

- ✓ C. Front cover:
 - 1. The removable front cover shall be mounted to the pump with 4 individual bolts.
 - 2. The front cover shall permit removal of the rotors without disturbing piping, bearings, and mechanical seals.
 - 3. The front cover shall be machined to accept a reversible wear plate.
- ✓ D. End cover:
 - 1. The removable end cover shall be flush with no recesses or dead pockets where solids can accumulate.
 - 2. The end cover shall be sealed with Buna-N O-ring and provide complete access to the pump chamber without disconnecting pipe work glands or bearings.
- ✓ E. Wear plates: Wear plates shall be constructed of Hardox 500 material, or equal, with a minimum Brinell hardness of 550 and a finished, hardened, reversible surface of 700 Brinell.
- ✓ F. Port connections: Provide ANSI Class 150 raised face flanges. Connections shall be suitable for field coating as specified in Section 09960 - High-Performance Coatings.

2.06 PUMP ROTORS

- ✓ A. Materials:
 - 1. Rotor core: As scheduled.
 - 2. Rotor (and/or rotor tip) coating: As scheduled.
- ✓ B. Construction:
 - 1. The pump shall utilize 2 tri or quad lobe rotors, which are driven through positive timing gears running in oil.
 - 2. Rotor cores shall be covered with a rotor coating as scheduled.
 - 3. The geometry of the rotor core shall be the same as that of the finished rotor.
 - 4. Rotor vane geometry shall be convoluted to provide pressure-pulse free operation.
 - 5. Designs with rotor vanes parallel to the shaft centerline will not be accepted.
 - 6. The convoluted rotor shall be specifically designed for pumping sludge containing organic solids, small inorganic particles, and grit.
 - 7. Rotors shall be positioned on the shaft by replaceable hardened key ways and secured to the shaft by internal/external expansion clamp sleeves and flush discs requiring no recesses in the end cover.
- ✓ C. In lieu of providing solid rotors, rotors may be provided with replaceable tips. Rotor, rotor tip, and rotor coating material shall be as specified.

2.07 SHAFTS

- ✓ A. Materials:
 - 1. Shaft: As scheduled.
 - 2. Shaft sleeve: As scheduled.



- B. Construction: Pump shafts shall be designed to provide sufficient stiffness to operate without distortion, damaging vibration, or excessive wear throughout the range of operation specified.



- C. Sludge wetted rotor/shaft connections are not acceptable.

Exception. This feature is unique to Vogelsang (only). Boerger design utilizes a mechanical seal chamber with an oil bath chamber which is vented to atmosphere; which allows instant verification of a mechanical seal leak.

2.08 STUFFING BOXES



- A. Construction:
 1. A blocking chamber located behind the mechanical seal and in front of the bearing housing lip seal shall be molded into the casting of the pump.
 - a. This chamber shall be suitable for oil fill through the top and bottom of the pump.
 - b. This chamber will have an external pressurized oil bottle mounted above the pump from the fill nipples to view the status of the mechanical seals.
 2. The external oil bottle will be located in view of the operator. Oil-quench shall provide lubrication and cooling of the seal, allow detection of seal failures, and provide a buffer zone to the sealed timing gear. Seal water flush systems are not acceptable.
 - a. In lieu of providing an external pressurized oil bottle mounted above the pump, manufacturer may provide a plastic stopper on the chamber that is vented to the atmosphere, which will allow fluid to escape as an indication of seal failure.



- B. Shaft seal type: As scheduled.



- C. Design of the pump shall allow removal and replacement of the seal via the front cover:
 1. Seal designs that open during rotor replacement are not acceptable.



- D. Oil drain gearbox and intermediate chamber shall be easily accessible:
 1. Oil drain under the pump is not acceptable.

2.09 GEAR REDUCERS (IF REQUIRED)



- A. If required for specified and scheduled pump configuration, gear reducers and couplings shall meet the requirements as specified in Section 15050 - Common Work Results for Mechanical Equipment.



- B. Provide NEMA C face connection between motor and gearbox.



- C. Provide helical reduction gears, rated for AGMA Class II service with a 1.5 service factor.



- D. Provide oil bath lubrication.

2.10 SUPPORTS, PEDESTALS, AND BASEPLATES



- A. Materials: Same as pump casing and coated as specified in Section 09960 – High-Performance Coatings or ASTM A283 steel, hot-dip galvanized after fabrication.

Anchor
Bolts
provided
by
others

- ✓ B. Pump, driver, and intermediate bearing support strength: Able to withstand minimum 1.5 times maximum imposed operating loads or imposed seismic loads, whichever is greater.
- ✓ C. Configuration:
 - 1. Support pump, gear reducers, and motor on a common structural steel baseplate.
 - 2. Pumping unit shall be furnished in a piggyback arrangement, belt driven with motor overtop of pump.
 - 3. Belt drive shall be as specified in Section 15050 - Common Work Results for Mechanical Equipment.
-  D. Anchor bolts: Designed by the pump manufacturer:
 - 1. 3/4-inch minimum diameter.
 - 2. As specified in Section 05190 - Mechanical Anchoring and Fastening to Concrete and Masonry.

2.11 EQUIPMENT GUARDS

- ✓ A. Provide safety guards as specified in Section 15050 - Common Work Results for Mechanical Equipment.

2.12 DRIVERS

- ✓ A. Horsepower:
 - 1. As scheduled.
 - 2. Listed driver horsepower is the minimum to be supplied:
 - a. Increase driver horsepower if required to prevent driver overload while operating at any point of the supplied pump operating head-flow curve including runout.
 - b. When scheduled driver is a motor, increase motor horsepower if required to prevent operation in the service factor.
 - c. Make all structural, mechanical, and electrical changes required to accommodate increased horsepower.
- ✓ B. Motors: Provide motors as specified in Section 16222 - Low Voltage Motors up to 500 Horsepower and as specified in this Section:
 - 1. Revolutions per minute: As scheduled.
 - 2. Enclosure: As scheduled.
 - 3. Electrical characteristics: As scheduled.
 - 4. Efficiency, service factor, insulation, and other motor characteristics: As specified in Section 16222 - Low Voltage Motors up to 500 Horsepower.
 - 5. Motor accessories: As specified in Section 16222 - Low Voltage Motors up to 500 Horsepower and in this Section and as indicated on the Drawings.
 - 6. Coordinate motors with the variable frequency drive manufacturer to ensure compatibility between the motor and variable frequency drive.

- ✓ C. Other drivers: As scheduled and as specified in sections listed in the Schedule.
- X D. Non-reverse ratchets: When scheduled, provide driver with non-reverse ratchets or pin mechanism to prevent reverse rotation of the pump and driver in the event of discharge valve failure.

Exception/Adder. A ratcheting motor specified. We don't see the need for this, and also the spec says it applies to vertical motors only. As our motors are horizontal, we will not be supplying a motor brake.

2.13 SPARE PARTS AND SPECIAL TOOLS

- ✓ A. Spare parts: Deliver the following for each type or size of pump:
 - 1. 1 spare rotor set or set of replaceable rotor tips.
 - 2. 1 set of mechanical seals.
 - 3. 1 set of O-rings.
 - 4. 1 set of wear plates (front, back, and radial).
- ✓ B. Special tools: For each type or size of pump specified, provide 1 set of all special tools required for complete assembly or disassembly of the pump system components.

PART 3 EXECUTION

3.01 COMMISSIONING

- ✓ A. As specified in Section 01756 - Commissioning and this Section.
- ✓ B. Manufacturer services:
 - 1. Provide certificates:
 - a. Manufacturer's Certificate of Source Testing.
 - b. Manufacturer's Certificate of Installation and Functionality Compliance.
 - 2. Manufacturer's Representative onsite requirements: Require manufacturer's representative to perform the following services as described below and as specified in Section 01756 - Commissioning. The specified durations are the minimum required time on the job site. Additional services and/or longer durations shall be provided as needed at no cost to the Owner to meet the required quality of work. Work to be done in a minimum of 4 trips:
 - a. Installation assistance: As required:
 - 1) Advise/observe the Contractor on the installation of the equipment.
 - 2) Provide additional assistance as required.
 - b. Installation inspection: 1 trip, 2-day minimum.
 - c. Functional Testing: 1 trip, 2-day minimum.
 - 3. Training: As defined in Section 01756 - Commissioning and this Section:
 - a. Operations: 2-hour class, 2 sessions.
 - b. Mechanical Maintenance: 3-hour class, 1 session.
 - a. Combined Electrical and I&C Maintenance: 1-hour class, 1 session.
 - 4. Final acceptance checkout: 1 workday (trip may be combined with training).
- ✓ C. Source testing: As specified in Pump Schedule. Test as specified in Section 15958 - Mechanical Equipment Testing.

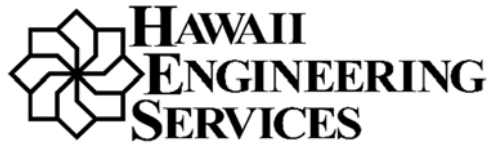
- ✓ D. Functional testing: As specified in Pump Schedule. Test as specified in Section 15958 - Mechanical Equipment Testing.
- ✓ E. Refer to Section 16950 - Field Electrical Acceptance Tests and Section 17950 - Commissioning for Instrumentation and Controls for additional requirements.

✓3.02 PUMP SCHEDULE

Tag Numbers	10-PMP-2210 10-PMP-2220
<u>General Characteristics:</u>	
Service	Blended Sludge
Name	Thickener Feed Pumps
Quantity	2
First Named Manufacturer's Model Number	Boerger, CL 390
Maximum Noise, dBA at 3 feet	85
Minimum Pump Displacement (gal/100 rev)	1,030
Minimum Pumped Fluid degrees Fahrenheit	50
Normal Pumped Fluid degrees Fahrenheit	70
Maximum Pumped Fluid degrees Fahrenheit	85
Vibration Analysis (Rotodynamic) Level	Not Required
Vibration Analysis Expert	Not Required
<u>Pump Characteristics:</u>	
Impeller Type	Positive Displacement, Rotary Lobe
Minimum Sphere Size, Inch	1.5
Shaft Seal Type	Single Mechanical Seal – Cartridge Type
Coupling Type	Spacer couplings
Speed Control	Variable Frequency Drive
Maximum Pump rpm	249
Minimum Pump rpm	61
Suction Flange Diameter, Inches	6
Discharge Flange Diameter, Inches	6
<u>Rated Design Point (at Maximum Revolutions per Minute):</u>	
Flow, gallons per minute	200
Head, psi	43
<u>Other Design Point:</u>	
Flow, gallons per minute	40

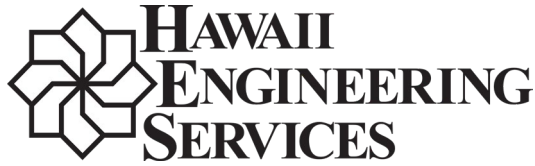
Tag Numbers	10-PMP-2210 10-PMP-2220
Head Range, psi	10
Maximum NPSHr at Every Specified Flow, feet	5.2
<u>Pump Materials:</u>	
Casing	Cast Iron
Top and Bottom Pump Housing	Grey Cast Iron
Rotor Core	Grey Cast Iron
Rotor Coating (including rotor tip coating)	Buna-N
Shaft	Carbon Steel
Shaft Sleeve	316 Stainless Steel
Nuts and Bolts	316 Stainless Steel
<u>Driver Characteristics:</u>	
Driver Type	Motor
Drive Arrangement	Belt Drive/Piggyback
Minimum Driver Horsepower	15
Maximum Driver rpm	1,800
<u>Motor Characteristics (when motor is driver type):</u>	
Inverter Duty Rated	Yes
Motor Voltage/Phases/Hertz	460/3/60
Enclosure Type	TEFC
<u>Source Testing:</u>	
Test Witnessing	Not Witnessed
Performance Test Level	2
Vibration Test Level	1
Noise Test Level	1
<u>Functional Testing:</u>	
Performance Test Level	2
Vibration Test Level	1
Noise Test Level	1

END OF SECTION



SUBMITTAL

PROJECT INFORMATION	
Submittal No.: S-0443 Job Name: Hilo Borger Rotary Lobe Pumps Location: Hilo Quote No: MM10163 Purchase Order No:	
CONTRACTOR	
Contractor: NAN INC Project Manager: Phone:	Address: 636 LAUMAKA ST HONOLULU, HI 96819-2312
MANUFACTURER'S REP	
Supplier: Project Manager: Rick Pitts Phone: 808-766-7288	Address: 1082 Makepono St
Transmittal	
☒ FOR REVIEW AND APPROVAL ☐ APPROVED ☐ REJECTED ☐ REVISE AND RESUBMIT ☐ ACKNOWLEDGE RECEIPT ☐ SEE REVIEW COMMENTS/RESPONSE BY: Rick Pitts DATE: September 2, 2025 ***NOTE*** On approval orders only, written submittal approval must be received before production can be authorized.	



1082 Makepono St. Honolulu, HI 96819 Ph:(808)841-0033 Fax:(808)841-2534 info@hiengineering.com

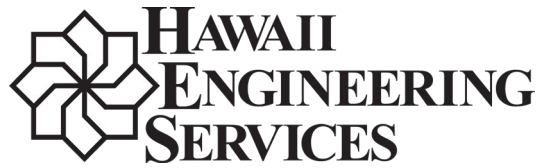
HILO WWTP Rehab and Replacement

Deviations/Exceptions to Specifications

Item	Section	Comments
	11312R	Single-Lobe Rotary Pumps
	1.05.C	Vibration Analysis and Rotodynamic analysis provided by others
	2.05.A.2	Boerger utilizes XAR 400; which has a Brinell Hardness rating of 370-430
	2.05.B	This is a carry-over from a specification written for a Vogelsang (only) pump. Boerger addresses this with a superior design feature whereas the wear liner itself is replaceable, thus no abrasive wear occurs on the pump housing itself, extending the useable life of the pump housing over the Vogelsang design, whereas abrasive wear occurs on the housing itself.
	2.05.E	Boerger utilizes XAR 400; which has a Brinell Hardness rating of 370-430
	2.08.A	This feature is unique to Vogelsang (only). Boerger design utilizes a mechanical seal chamber with an oil bath chamber which is vented to atmosphere; which allows instant verification of a mechanical seal leak.
	2.10.D	Anchor Bolts provided by others
	2.12.D	Exception/Adder. A ratcheting motor specified. We don't see the need for this, and also the spec says it applies to vertical motors only. As our motors are horizontal, we will not be supplying a motor brake.
	3.02	Clarification. Pump performance table suggests an NPSHr of 5.2 ft. This is unusual, as the norm is the consulting engineer specifies the NPSHa, then the pump manufacturer calculates the actual, or the NPSHa. We then recalculate and make sure the NPSHa is > the NPSHr. Please provide NPSHa.

1.05.C Further Clarification:

- Vibration testing is more commonly performed on rotating equipment that runs at much higher RPM's (thousands of RPM) vs a RL pump which spins at more like 200 RPM. As a result, the



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National Hydraulic Institute does not have any set standards for RL pumps. Therefore, there is no set benchmark established for pass/fail criterion. This brings the need for such a test on an RL into question. All we can provide is the actual measurement that it tests at and provide that data. We are not familiar with any WW plants that measure vibration on their RL pumps as a means of reliability or preventative maintenance. We suggest the request for this test be deleted.

- If the intent is for field testing, then this would have to be performed “by others”.

TS25402124

Technical Submittal

Project: Hilo WWTP

Equipment: Rotary Lobe Pumps

Specification: 11312R

Description:	Rotary Lobe Pumps
Model:	CL 390
Tags:	10-PMP-2210 / 10-PMP-2220
Quantity:	2

Preface

The following submittal has been compiled with the sole intent of providing our client with a sufficiently detailed account of the technical aspects of the equipment proposed in order that they may be satisfied that the units meet the requirements of the reference applications.

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8. *Spare Parts Supplied*
9. *Dry Run Protection*
10. *Factory Testing*
11. *Anchor Bolts*
12. *Coatings*
13. *Warranty Certificate*

Pump Data Sheet

BORGER ROTARY LOBE PUMPS

ROBUST, RELIABLE AND CUSTOMIZED

Börger rotary lobe pumps have been used in the municipal and environmental technology sector for more than 25 years. The solids handling pumps are powerful, reliable and efficient. Börger pumps convey dirty, sludgy and abrasive media without problems.



OPERATING PRINCIPLE

Börger rotary lobe pumps are self-priming, valveless, positive displacement pumps. The synchronized rotation of the rotor pair creates a vacuum on the priming side of the pump. This vacuum draws the liquid into the pump chamber. The medium is pumped into the pressure area due to the rotation of the rotors. If the direction of rotation is changed, the flow is reversed.

CONSTRUCTION

- 1 The quick-release cover**
Access to all wetted parts by simply loosening four ring nuts.
- 2 The rotors**
Large selection of high-quality rotors for almost pulsation-free pumping of the medium.
- 3 The casing protection**
The casing liners and casing protection plates protect the pump casing from wear and can be replaced in a matter of minutes.
- 4 The intermediate chamber and shaft seal**
The large-volume quench chamber is oil-filled and ensures the highest degree of safety. The ideal mechanical seal is selected depending on the medium.
- 5 The carrier and timing gear**
The high-quality and maintenance-free carrier and timing gear guarantees a smooth and even operation of the rotors. The result is a long service life of the pump.



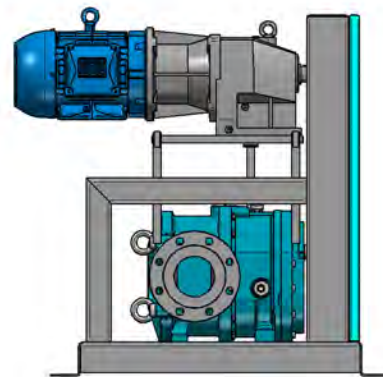
Structure



Pump feet indicate
mounting position

1. Timing gear oil fill
2. Intermediate chamber oil fill
3. Oil sight glass
4. Timing & Intermediate chamber drain
5. Corrosion resistant nameplate

Overhead Configuration Supplied



Boerger, LLC
2860 Water Tower Place
Chanhassen, MN 55317

Tel.: 612-435-7300
Fax: 612-435-7301

Customer: Hilo WWTP
Pump Type: Classic CL 390
Serial No.: TBD
Code: PC3SARCF AAAAGCCC16Z
S.O. No. 25402124

Tags: 10-PMP-2210
10-PMP-2220

Delivery Date:

Conveying Product		Operating Data	
Name	Blended Sludge	Location	Closed building, dry
Viscosity	cP	Hazardous Area	Not Classified
Solid Content	-	Environmental Temp.	Ambient °F
Solid Size	-	Operating Mode	Not Classified
Density/Specific Gravity	-	Suction Pressure	Flooded
Temperature	50 - 85 °F		
ph-value	Neutral		
Performance		Flow	Discharge Pressure
Min. Capacity	40.0 gpm	10.0 psi	61 rpm
Norm. Capacity	200.0 gpm	43.0 psi	249 rpm
Max. Capacity			
Code	Decoding		
1. - 3.	PC3	Pump Type	Classic CL 390
4.	S	Gear Design (Bearings)	Standard
5.	A	Drive Shaft Design	1x drive shaft, cylindrical with keyway, pos. 0, steel 1.7225/AISI 4140H
6.	R	Flow Direction	Reversible
7.	C	Lipseals / Inner Ring	Classic NBR / steel 1.3505/AISI 52100
8.	F	Pump Casing	Grey Cast iron 0.6025/ASTM A48-40B, with radial MIP liners
9.	A	Cover & Axial Plates	Hardened steel 1.8714/1.7225/ AISI 4140H
10.	A	Radial Liners	Hardened steel 1.8714/1.7225/ AISI 4140H
11.	A	Holding Bushings	Steel, Standard
12.	A	Mechanical Seal	Mechanical Seal, Seal Rings: Chill Casting, O-rings: NBR
13.	G	Rotor Design	Tri-lobe, Screw Profile, MIP Lobe Tips
14.	C	Rotor / Tip Material	NBR
15.	C	O-Rings (static)	NBR
16.	C	Flange Gaskets	NBR
17.	1	Mounting Position	Pump vertically, Feet down (M1)
18.	6	Series Index	6
19.	Z	Special Equipment	Temperature sensor in pump cover
Lubricants		Drive Data	
Intermediate Chamber	Mobil DTE 10 Excel 68	Make / Model	WEG Z01518ET3E254TC
Timing Gear	Mobilgear 600 XP 220	Power	15 Hp
Flanges - B6		Speed	1765 rpm
Material	316 SS Steel	Phase/Hz/Voltage	3/60/(208-230/460)
Suction Side	6" ANSI 150-lb	Enclosure	TEFC
Pressure Side	6" ANSI 150-lb	Option	Space Heater / Thermostats
Drive Connection		Option	AEGIS Grounding Ring / Routine Test
Drive Belt	Synchronous Drive Belt	Serial Number	
Sprockets		Gear Reduction	
Base Frame	Galvanized Steel	Make / Model	Nord SK42VL-AN250TC
Configuration	Overhead	Output Speed	283 rpm
Gearing		Gear Ratio	6.19:1
L-10 (B-10) Bearing Life	100,000 h	Option	
		Serial Number	
Options: *** Motor options continued - SS Breather Drains & SS Hardware			
Other:			

Construction



1. Quick release cover – Allows access to wet end components, without removing pump from piping.
2. Axial liners – Hardened steel, 550 Brinell
3. Tri-Lobe screw Rotor with replaceable tips – ASTM A536 core with fully encapsulated Buna-N (NBR) tips
4. Mechanical Seals: Chill-casting (duronite) face with Buna-N O-rings
5. Radial liners – Hardened steel, 550 Brinell
6. Single Block Casing – Grey Cast Iron, ASTM A48-40 B



**** Rotor orientation front view**
**** Removable tips front view**

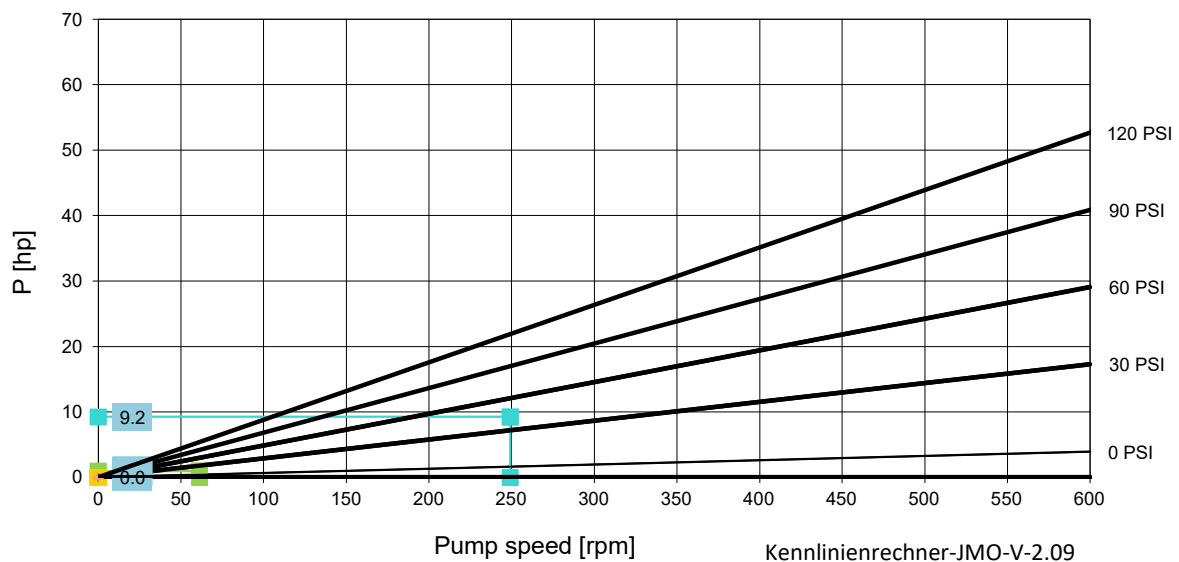
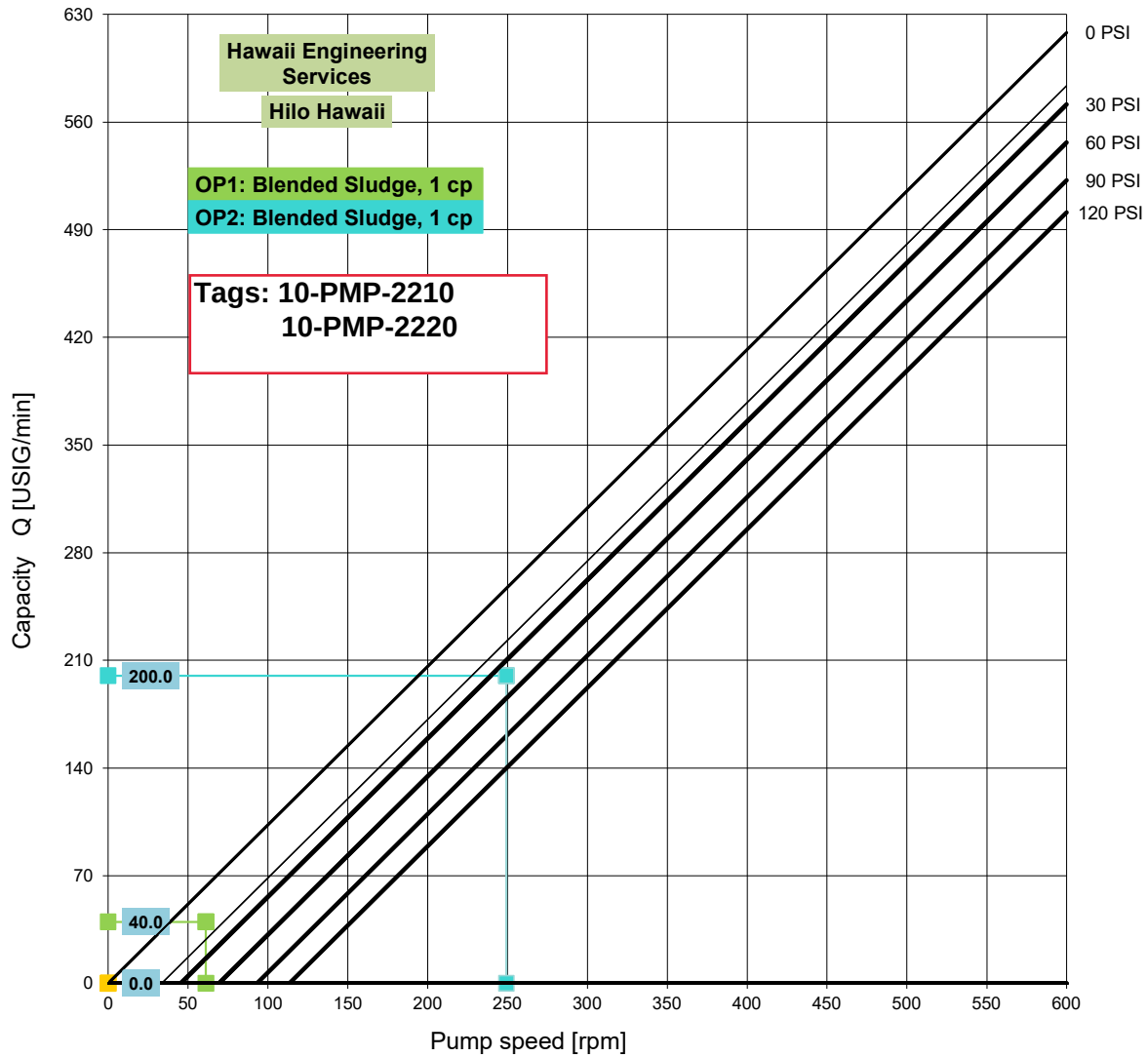


Pump Performance Curve

Performance Curve CL390 Rotor Type G



BÖRRER®



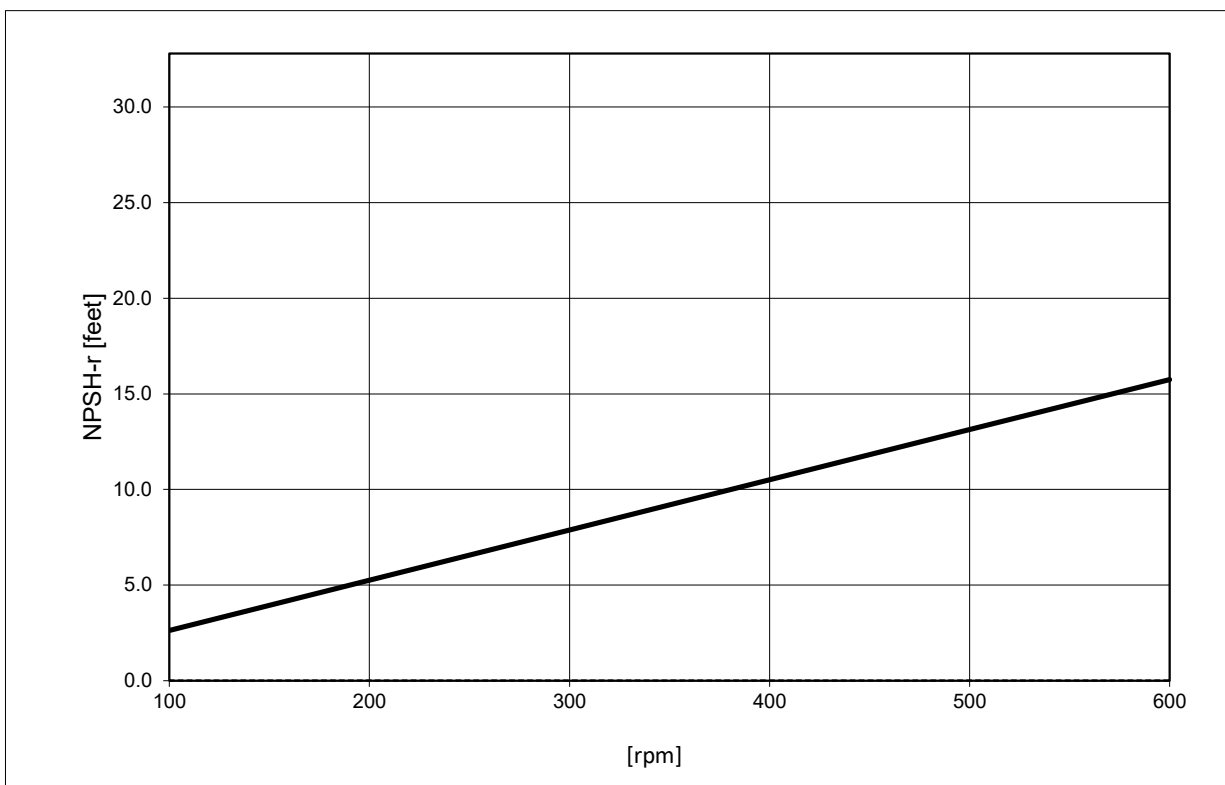
NPSH-Performance Curve



BÖRRER®

CL390	G	Kennlinienrechner-JMO-V-2.09
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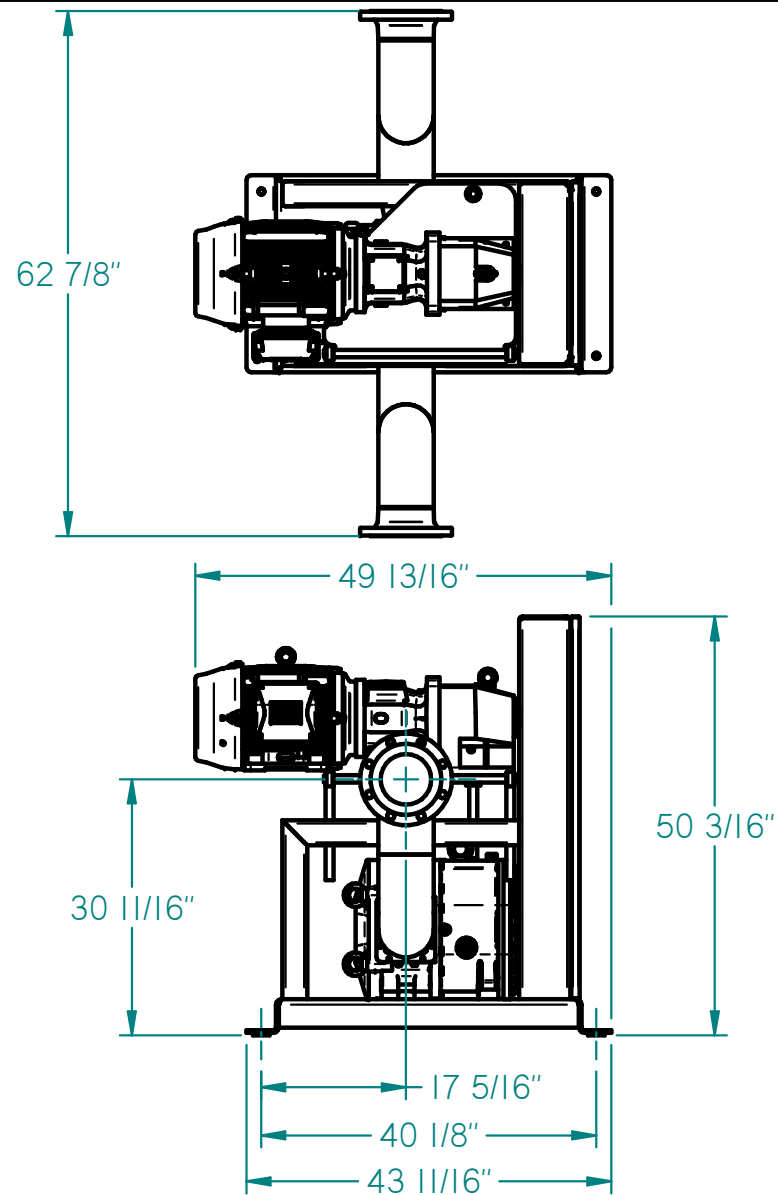
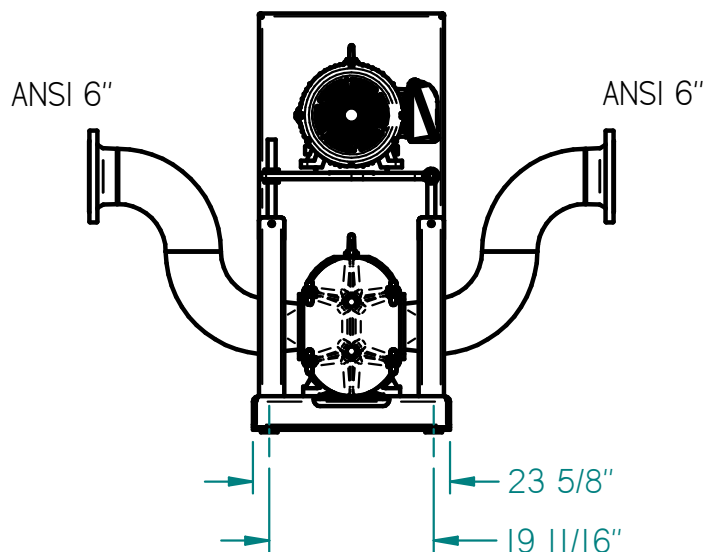
n [rpm]	100	200	300	400	500	600
NPSH-r [feet]	2.6	5.2	7.9	10.5	13.1	15.7
NPSH-a [feet] OP1= 32.8	30.2	27.6	24.9	22.3	19.7	17.1
NPSH-a [feet] OP1= 32.8	30.2	27.6	24.9	22.3	19.7	17.1
NPSH-a [feet] OP1= 10						



NPSH-a	Fluid pressure at pump inlet in meter water column [mws]
NPSH-r	Decrease of pressure due to fluid acceleration inside pump [mws]
NPSH-a > NPSH-r Delta > 0	Pump runs smooth and works with design flow
NPSH-a < NPSH-r Delta < 0	Runs with noise flow is beneath design flow - short-term operation allowed
NPSH-a < 0	flow of fluid impossible

GA Drawings

Tags: 10-PMP-2210
10-PMP-2220



NOTE:
I. TOTAL WEIGHT = 1,373 LBS

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BOERGER, LLC

2860 Water Tower Place
Chanhassen, MN 55317
TEL: 612-435-7300
FAX: 612-435-7301



BÖRGER

Model: CL 390
Gear: Nord - In-Line SK42-ANVL-AN250TC
Power: 15 HP

ALL DIMENSIONS IN INCHES
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Belt Drive Data Sheets

HT500 synchronous belt drives

Reduce maintenance. Improve reliability. Save money.

With 140 years of producing mechanical belt drive products, Dodge is an industry leader. HT500 is the latest generation of high-torque synchronous belt drive systems. The innovative HT500 synchronous belt and sprocket system requires less re-tensioning than V-belts and no lubrication. The result: maintenance-free power transmission.



Keep employees safe

- Downsize large, heavy V-belt drives by up to 50 percent with compact, lighter-weight HT500 synchronous drives, and make installations easier and safer
- Reduce the risk of slipping on grease or oil in workspace area by replacing roller chain with a lubrication-free belt system

Reduce total cost

- Eliminate V-belt slip and improve energy efficiency to as high as 98 percent with a positively engaged synchronous drive
- Get more torque from a less-bulky drive and save money upfront with an economical, compact HT500 selection

Extend equipment life

- The design and construction of HT500 can handle more torque than traditional V-belts or synchronous belts. Go longer between belt replacements with HT500 and improve the reliability of your drive
- Synchronous belts require less re-tensioning than V-belts. When properly installed synchronous drives can reduce overhung load on connected equipment, improve reliability, and keep applications running longer

Set up your account at ptwizard.com

- Build your packages step-by-step
- View automated quotes for your selected items
- Review product technical details and confirm selection



The HT500 synchronous belt drive product line is one of many examples of the Dodge commitment to customer success. Contact sales or engineering for additional supplier service and support.



• Mechanically retained flanges

Dodge synchronous flanges are mechanically retained to ensure no staking or welding of the flanges is required. Flanges can withstand up to 1000 pounds of force before removal from the sprocket

• Modified-curvilinear belt teeth

Fewer fatigue points, less wear, less sliding, less noise, low backlash, and a generally more effective design than other synchronous belt tooth profiles, such as trapezoidal, parabolic, or curvilinear toothed belts

• Carbon fiber cords

Prevents elongation, a common issue in V-belts and roller chain. Carbon fiber is also capable of higher torque transmission than other common belt cord materials, such as polyester and fiberglass

HT500 belt and sprocket sizes

Belt width	Sprocket tooth range	Bushing min - max	Belt length range
8 mm pitch			
12mm	22S – 224S	MPB/1008 – 2517	248MM – 4480MM
21mm	22S – 224S	MPB/1008 – 3020	248MM – 4480MM
36mm	22S – 224S	MPB – 3525	248MM – 4480MM
62mm	22S – 224S	MPB – 3526	640MM – 4480MM
14mm pitch			
20mm	28S – 224S	2012 – 4030	994MM – 4410MM
37mm	28S – 224S	MPB/2012 – 4030	994MM – 4410MM
68mm	28S – 224S	MPB – 5040	994MM – 4410MM
90mm	28S – 224S	MPB – 6050	994MM – 4410MM
125mm	28S – 224S	MPB – 7060	994MM – 4410MM

• Polyurethane and nylon

Constructed with a polyurethane backing and black nylon tooth face, the HT500 belt is resistant to wet, abrasive, and high-temp environments up to 185°F

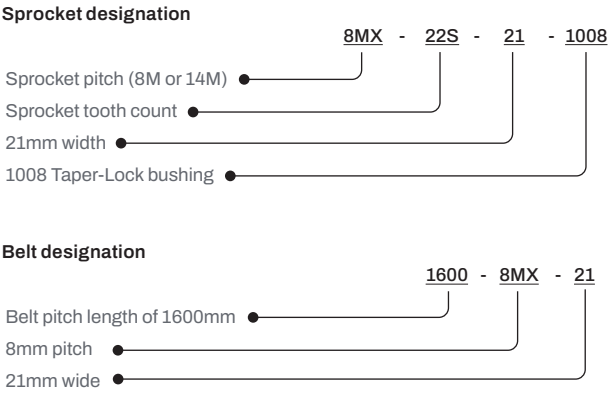
• Taper-Lock hubs

HT500 belts are more narrow than older generations of synchronous belts. With Taper-Lock bushings and sprocket hubs, the HT500 package is an extremely compact design. A smaller, lighter assembly is easier to handle during installation and can reduce the overhung load on connected equipment

• Designed to ARPM/MPTA standards

Rugged and reliable sprockets manufactured to the highest quality standards. HT500 sprockets are also protected by corrosion-resistant coatings, including E-coat, making them more resistant to rust and paint chipping than any other line of sprockets on the market

HT500 part nomenclature:



Gear Reducer Data Sheets

HELICAL IN-LINE

SPEED REDUCERS/GEAR MOTORS

PERFORMANCE SPECIFICATIONS

Configuration:concentric

Motor HP (min/ max) : 0.16 / 200

Motor kW (min/ max) : 0.12 / 160

Typical efficiency : 97.5%

Number of gear reductions : 1 to 6

MOUNTING STYLES

Footed housing style:standard

B5 flange outside diameter range [in] : 4.72 to 21.65

B5 flange outside diameter range [mm] : 120 to 550

B14 flange outside diameter range [in] : 3.54 to 7.87

B14 flange outside diameter range [mm] : 90 to 200

OPTIONS

NEMA C-face or IEC input

Custom adapter flange

Flange pilot removed

RATIO AND SPEED

Minimum standard ratio : 1.24 : 1

Maximum standard ratio : 13304.45 : 1 (compound gear unit)

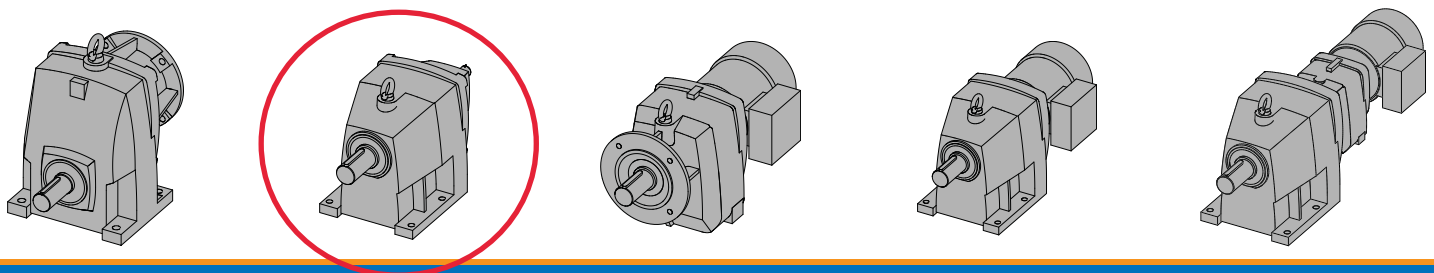
Minimum output speed from 1750 rpm motor : 0.13 rpm

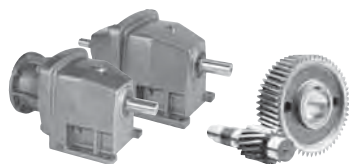
Maximum output speed from 1750 rpm motor : 1411 rpm



Unit Size	Torque Max*		Ratio Range	Shaft Diameter		Unit Size	Torque Max*		Ratio Range	Shaft Diameter	
	[lb-in]	[Nm]		[in]	[mm]		[lb-in]	[Nm]		[in]	[mm]
SK 02	876	99	2.95 - 73.06	0.750	20	SK 51E	4,354	492	1.24 - 13.27	1.625	40
SK 03	974	110	65.50 - 313.11	0.750	20	SK 52	17,912	2,024	2.78 - 86.92	2.250	55
SK 11E	513	58	1.35 - 9.11	0.750	20	SK 53	19,744	2,231	58.94 - 728.20	2.250	55
SK 12	1,628	184	2.96 - 72.63	1.000	25	SK 62	27,612	3,120	2.97 - 48.73	2.500	65
SK 13	1,735	196	68.40 - 420.83	1.000	25	SK 63	32,745	3,700	17.37 - 372.21	2.500	65
SK 21E	681	77	1.46 - 10.20	1.000	25	SK 72	41,666	4,708	2.76 - 43.71	3.000	75
SK 22	3,310	374	2.79 - 86.30	1.250	30	SK 73	50,003	5,560	18.00 - 205.61	3.000	75
SK 23	3,009	340	64.80 - 516.65	1.250	30	SK 82	64,127	7,246	2.89 - 48.82	3.500	90
SK 31E	1,637	185	1.39 - 10.20	1.250	30	SK 83	81,243	9,180	21.04 - 216.61	3.500	90
SK 32	6,284	710	2.96 - 81.27	1.625	40	SK 92	93,925	10,612	3.51 - 35.47	4.250	110
SK 33N	5,947	672	88.18 - 740.37	1.625	40	SK 93	123,900	14,000	19.12 - 187.89	4.250	110
SK 41E	2,567	290	1.41 - 14.80	1.375	35	SK 102	153,698	17,367	4.28 - 38.81	5.250	130
SK 42	11,009	1,244	3.02 - 105.08	1.875	45	SK 103	204,966	23,160	21.19 - 207.47	5.250	130
SK 43	11,408	1,289	40.98 - 1071.82	1.875	45						

* Torque may vary by ratio





Model Type	Gear Ratio	Output Speed n_2 1750 rpm [rpm]	Output Torque* $T_{2\max}$ [lb-in]	Maximum input power [◆] Solid input shafts type "W"				NEMA C-Face* Available Combinations							
				Input Speed				56C	140TC	180TC	210TC	250TC	280TC	320TC	360TC
				1750 rpm [hp]	1150 rpm [hp]	875 rpm [hp]	580 rpm [hp]								
SK 41E	1.41	1241	1593	20.00	13.20	10.00	6.60			X	X	X			
	1.50	1167	1682	20.00	13.20	10.00	6.60			X	X	X			
	1.63	1074	1770	20.00	13.20	10.00	6.60			X	X	X			
	1.82	962	1974	20.00	13.20	10.00	6.60	X	X	X	X	X			
	2.14	818	2195	20.00	13.20	10.00	6.60	X	X	X	X	X			
	2.50	700	2398	20.00	13.20	10.00	6.60	X	X	X	X	X			
	3.08	568	2567	20.00	13.20	10.00	6.60	X	X	X	X	X			
	3.42	512	1239	10.07	6.64	5.03	3.32	X	X	X	X				
	3.88	451	1283	9.18	6.06	4.59	3.03	X	X	X	X*				
	4.29	408	1372	8.88	5.86	4.44	2.93	X	X	X	X*				
	5.27	332	1726	9.09	6.00	4.55	3.00			X	X*				
	7.18	244	1682	6.51	4.30	3.25	2.15			X	X*				
	10.55	166	1682	4.43	2.92	2.21	1.46	X	X	X*					
	14.80	118	1177	2.20	1.45	1.10	0.73	X	X						
SK 42	3.02	579	5345	20.00	13.20	10.00	6.60			X	X	X			
	3.21	545	5487	20.00	13.20	10.00	6.60			X	X	X			
	3.50	500	5885	20.00	13.20	10.00	6.60			X	X	X			
	3.89	450	6195	20.00	13.20	10.00	6.60	X	X	X	X	X			
	4.58	382	6832	20.00	13.20	10.00	6.60	X	X	X	X	X			
	4.79	365	8496	20.00	13.20	10.00	6.60			X	X	X			
	5.10	343	8708	20.00	13.20	10.00	6.60			X	X	X			
	5.35	327	7230	20.00	13.20	10.00	6.60	X	X	X	X	X			
	5.75	304	10036	20.00	13.20	10.00	6.60			X	X	X			
	6.19	283	9514	20.00	13.20	10.00	6.60	X	X	X	X	X			
	6.65	263	10293	20.00	13.20	10.00	6.60			X	X	X			
	7.28	240	9523	20.00	13.20	10.00	6.60	X	X	X	X	X			
	8.50	206	9523	20.00	13.20	10.00	6.60	X	X	X	X	X			
	10.20	172	10328	20.00	13.20	10.00	6.60	X	X	X	X	X			
	12.28	143	10585	20.00	13.20	10.00	6.60	X	X	X	X	X			
	14.38	122	10248	19.84	13.09	9.92	6.55	X	X	X	X	X*			
	15.12	116	11009	20.00	13.20	10.00	6.60	X	X	X	X	X			
	17.71	99	10496	16.49	10.88	8.24	5.44	X	X	X	X	X*			
	17.92	98	8832	13.73	9.06	6.87	4.53			X	X				
	21.50	81	10293	13.23	8.73	6.61	4.37			X	X				
	21.87	80	9700	12.31	8.13	6.16	4.06	X	X	X	X	X*			
	24.41	72	7593	8.67	5.73	4.34	2.86			X	X*				
	24.67	71	7885	8.88	5.86	4.44	2.93	X	X	X	X*				
	25.88	68	11001	11.87	7.83	5.93	3.92			X	X				
	29.29	60	9036	8.60	5.68	4.30	2.84			X	X*				
	30.46	57	9540	8.63	5.69	4.31	2.85	X	X	X	X*				
	35.25	50	10868	8.62	5.69	4.31	2.85			X	X*				
	41.29	42	10496	6.99	4.62	3.50	2.31			X	X*				
	50.98	34	9717	5.24	3.46	2.62	1.73			X	X*				
	60.66	29	8885	4.09	2.70	2.04	1.35	X	X	X*					
	74.87	23	9558	3.49	2.30	1.74	1.15	X	X	X*					
	85.10	21	7045	2.35	1.55	1.17	0.77	X	X						
	105.08	17	7629	2.06	1.36	1.03	0.68	X	X						

* Caution - The motor power may exceed the gear unit's mechanical torque capacity
 ◆ The mechanical power limit of the solid input shaft type "W" may limit the reducer rating.
 All ratings are mechanical. See page 14 for thermal considerations.

	W	56C	140TC	180TC	210TC	250TC
SK 41E	101	90	106	106	137	159
SK 42	143	132	148	148	179	201

Motor Data Sheets

Severe Duty is Standard with WEG W22 motors.

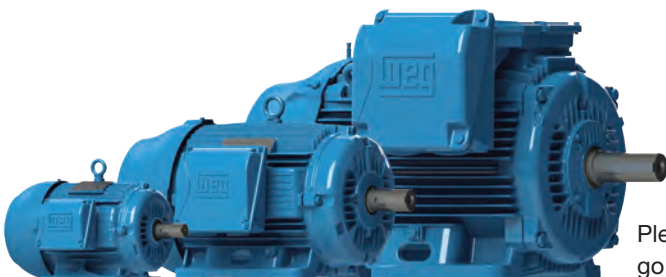
You do not need a special motor for severe duty. Severe Duty is standard with WEG W22 motors.

High performance with maximum energy efficiency is the goal of the WEG W22 electric motor. High efficiency and low cost of ownership throughout the entire motor lifetime have been the basis for the W22 development. A design created to maximize performance and energy savings. Optimized cooling system, large and accessible terminal box, exclusive bearing seal system, low vibration levels, and a 1.25 service factor are just a few of many STANDARD features of our W22 motors.

Available in NEMA Premium or Super Premium Efficiency levels.

No matter what you call it, WEG's W22 standard product is designed for:

DUTY	STANDARD
Mill-Chem	✓
Tough Service	✓
Severe Duty	✓
Crusher Duty	✓
WEG Duty	✓



Please contact your authorized WEG distributor or go to weg.net/us for further information or additional specifications. **1-800-ASK-4WEG**



Three Phase General Purpose Motors

W22 - Cast Iron

Three-Phase
General Purpose
Motors

Standard Features:

- 50/60 Hz rated (380V on nameplate - 400/415 available)
- Rated 50HZ at same horsepower up to 250HP 1.0 Service Factor for 2,4 & 6 pole motors
- Rated 50Hz at same horsepower up to 125HP 1.0 Service Factor for 8 pole motors
- Class F insulation (impregnation resin and magnet wire are class H)
- F1 mounted (cast iron frames are F2 convertible). Frames 447T and up can easily be F2 converted by simply rotating the terminal box adapter
- Cast iron frames: regreasable ball bearings (frames 254T and up)
- Gasketed conduit box

Inverter Rated

1000:1 for Variable Torque

NEMA Premium

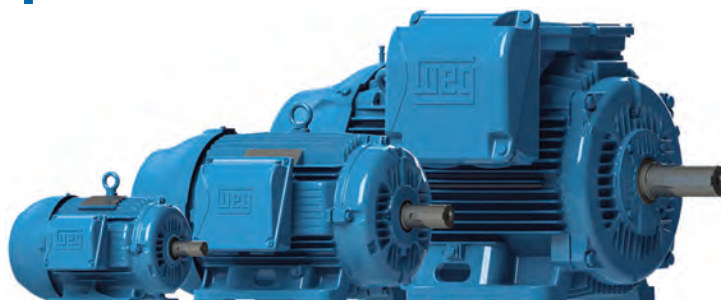
20:1 Constant Torque up to 250HP

6:1 Constant Torque 300HP and up

High Efficiency (EPAct)

12:1 Constant Torque up to 250HP

4:1 Constant Torque 300HP and up



Features that make a difference:

- All NEMA Premium ratings have a 1.25 service factor (up to 100 HP) resulting in cooler operation and extended life of the motor
- All Cast Iron Construction, including Terminal Box and Fan Cover (*)
- Solid feet for reduced vibration levels and impact absorption
- Optimized ventilation system for cooler operation and extended life
- High Grade FC200 cast iron provides superior mechanical strength and heat dissipation
- All WEG W22 motors are Totally Enclosed Fan Cooled with a true IP55 rating against dust and moisture. (IPW56, IPW65 and IPW66 available as optional)
- Exclusive W-Seal 364T and larger provides superior bearing protection
- Taconite Labyrinth seal 586 Frame and larger
- Exclusive WEG painting system exceed 200hrs ASTM 117 corrosion test (Exceeds IEEE841 standard)
- Balanced to 0.08 inches per second vibration limits (Meets IEEE841 standard)
- Four Bolt Conduit Cover with glued Neoprene Gasket
- Impregnation Resin and magnet wire are insulation class H
- Stainless Steel Nameplate - Laser etched with high contrast background
- Corrosion Proof Drains
- Inverter Duty per NEMA MG1, Part 31
- Certified Class I Div 2, Groups A, B, C & D; Class II, Div 2, Groups F & G
- Suitable for IEEE45 and USCG 259 – Marine Duty

*Cast iron fan cover available as an option on 143-215T frames

ADDITIONAL TECHNICAL DATA

ELECTRICAL DATA: p. B-6

MECHANICAL DATA: p. C-2



DATA SHEET

Three Phase Induction Motor - Squirrel Cage

Tags: 10-PMP-2210
10-PMP-2220



Customer :

Product line : W22 NEMA Premium Efficiency Three-Phase
Product code : 12446945
Catalog # : 01518ET3E254TC-W22

Frame	: 254/6TC	Locked rotor time	: 30s (cold) 17s (hot)
Output	: 15 HP (11 kW)	Temperature rise	: 80 K
Poles	: 4	Duty cycle	: Cont.(S1)
Frequency	: 60 Hz	Ambient temperature	: -20°C to +40°C
Rated voltage	: 208-230/460 V	Altitude	: 1000 m.a.s.l.
Rated current	: 39.8-36.0/18.0 A	Protection degree	: IP55
L. R. Amperes	: 255-230/115 A	Cooling method	: IC411 - TEFC
LRC	: 6.4x(Code G)	Mounting	: F-1
No load current	: 11.7-13.6/6.80 A	Rotation ¹	: Both (CW and CCW)
Rated speed	: 1765 rpm	Noise level ²	: 64.0 dB(A)
Slip	: 1.94 %	Starting method	: Direct On Line
Rated torque	: 44.6 ft.lb	Approx. weight ³	: 315 lb
Locked rotor torque	: 229 %		
Breakdown torque	: 270 %		
Insulation class	: F		
Service factor	: 1.25		
Moment of inertia (J)	: 2.62 sq.ft.lb		
Design	: B		

Output	50%	75%	100%	Foundation loads	
Efficiency (%)	91.0	91.7	92.4	Max. traction	: 489 lb
Power Factor	0.68	0.78	0.83	Max. compression	: 804 lb

	Drive end	Non drive end
Bearing type	: 6309 C3	6209 C3
Sealing	: V'Ring	V'Ring
Lubrication interval	: 20000 h	20000 h
Lubricant amount	: 13 g	9 g
Lubricant type	Mobil Polyrex EM	

Notes

Supplied with: Custom

- Space Heater 120V
- N/C Thermostats
- AEGIS Grounding Ring
- SS Breather Drains
- SS Hardware
- Routine Test

This revision replaces and cancel the previous one, which must be eliminated.

- (1) Looking the motor from the shaft end.
- (2) Measured at 1m and with tolerance of +3dB(A).
- (3) Approximate weight subject to changes after manufacturing process.
- (4) At 100% of full load.

These are average values based on tests with sinusoidal power supply, subject to the tolerances stipulated in NEMA MG-1.

Rev.	Changes Summary	Performed	Checked	Date
Performed by				
Checked by				
Date	26/02/2020			
			Page	Revision
			1 / 6	

TORQUE AND CURRENT VS SPEED CURVE

Three Phase Induction Motor - Squirrel Cage

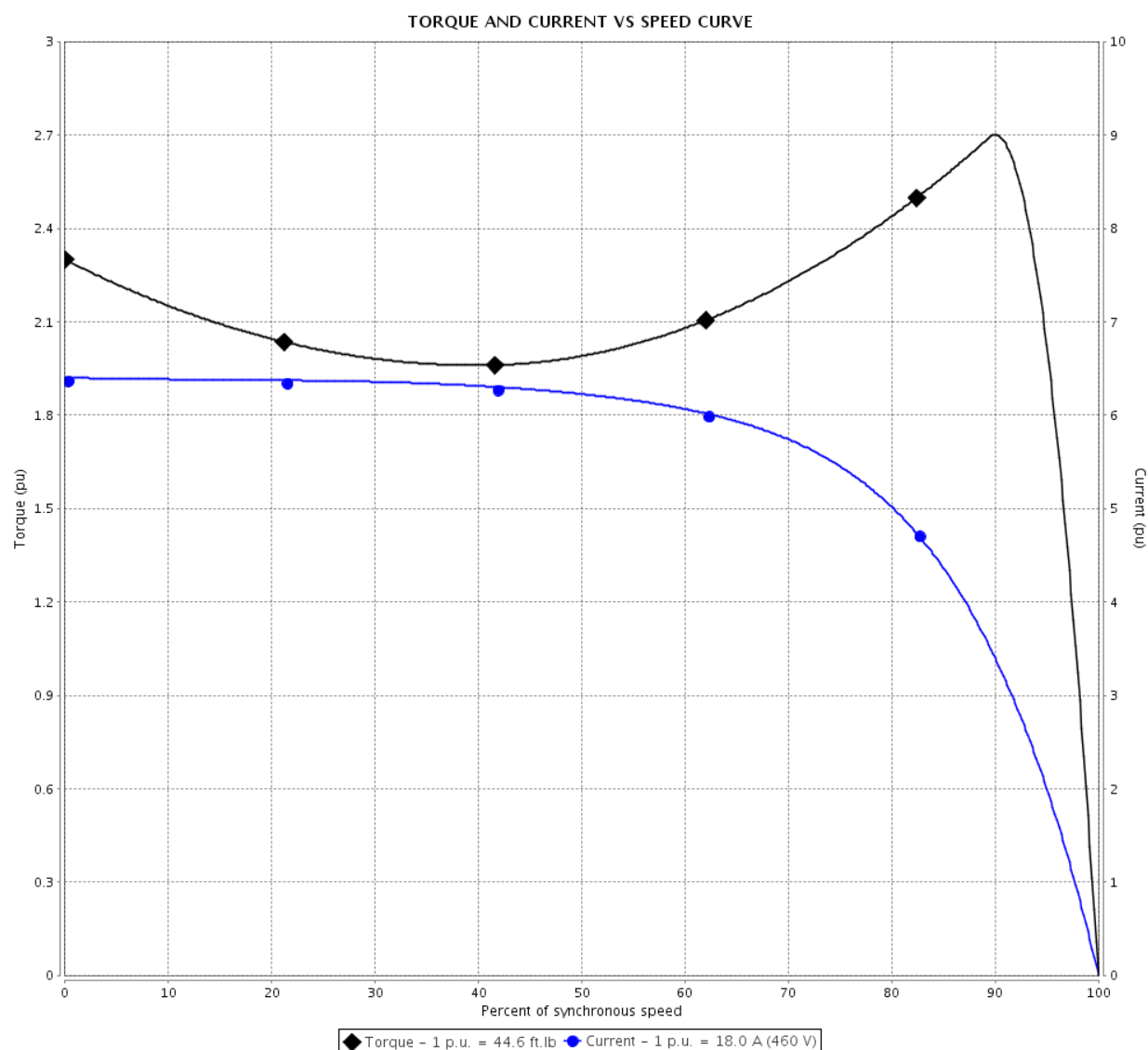


Customer :

Product line : W22 NEMA Premium Efficiency
Three-Phase

Product code : 12446945

Catalog # : 01518ET3E254TC-W22



Performance : 208-230/460 V 60 Hz 4P

Rated current : 39.8-36.0/18.0 A
LRC : 6.4
Rated torque : 44.6 ft.lb
Locked rotor torque : 229 %
Breakdown torque : 270 %
Rated speed : 1765 rpm

Moment of inertia (J) : 2.62 sq.ft.lb
Duty cycle : Cont.(S1)
Insulation class : F
Service factor : 1.25
Temperature rise : 80 K
Design : B

Locked rotor time : 30s (cold) 17s (hot)

Rev.	Changes Summary	Performed	Checked	Date
Performed by			Page	Revision
Checked by			2 / 6	
Date				

LOAD PERFORMANCE CURVE

Three Phase Induction Motor - Squirrel Cage

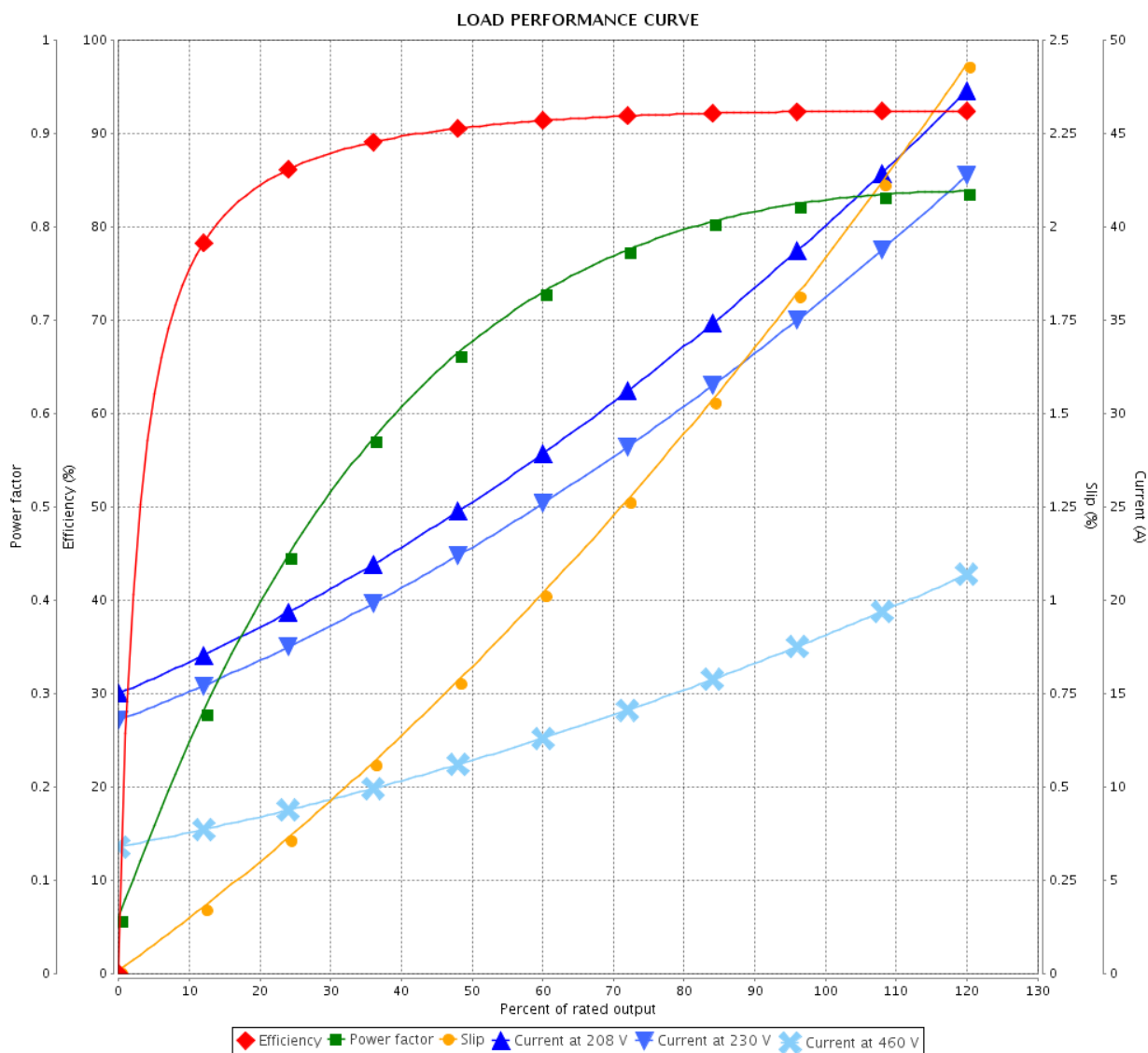


Customer :

Product line : W22 NEMA Premium Efficiency
Three-Phase

Product code : 12446945

Catalog # : 01518ET3E254TC-W22



Performance : 208-230/460 V 60 Hz 4P

Rated current : 39.8-36.0/18.0 A
LRC : 6.4
Rated torque : 44.6 ft.lb
Locked rotor torque : 229 %
Breakdown torque : 270 %
Rated speed : 1765 rpm

Moment of inertia (J) : 2.62 sq.ft.lb
Duty cycle : Cont.(S1)
Insulation class : F
Service factor : 1.25
Temperature rise : 80 K
Design : B

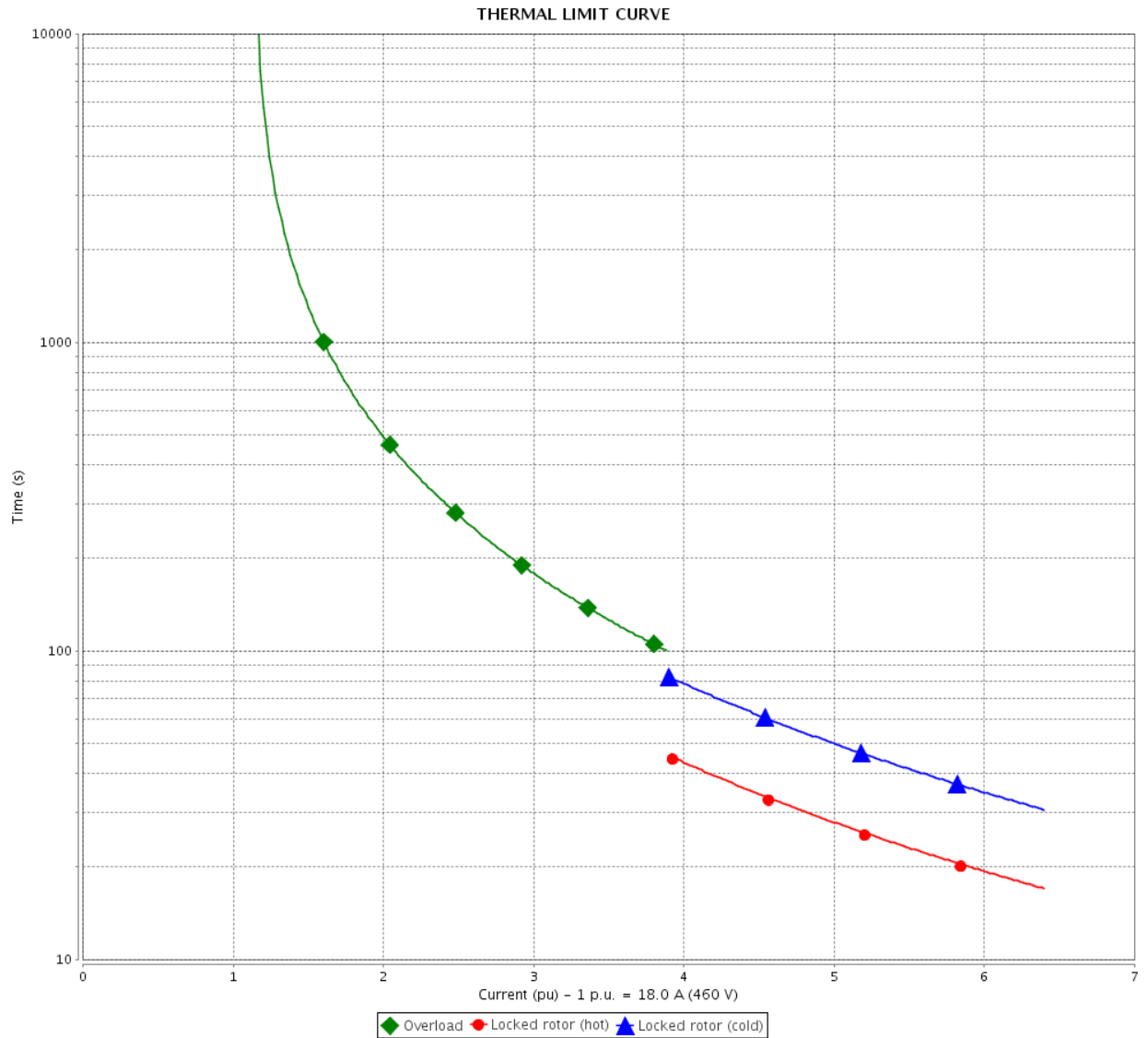
Rev.	Changes Summary	Performed	Checked	Date
Performed by		Page 3 / 6Revision		
Checked by				
Date	26/02/2020			

THERMAL LIMIT CURVE

Three Phase Induction Motor - Squirrel Cage



Customer : _____



Rev.	Changes Summary		Performed	Checked	Date
Performed by				Page 5 / 6	Revision
Checked by					
Date	26/02/2020				

VFD OPERATION CURVE

Three Phase Induction Motor - Squirrel Cage

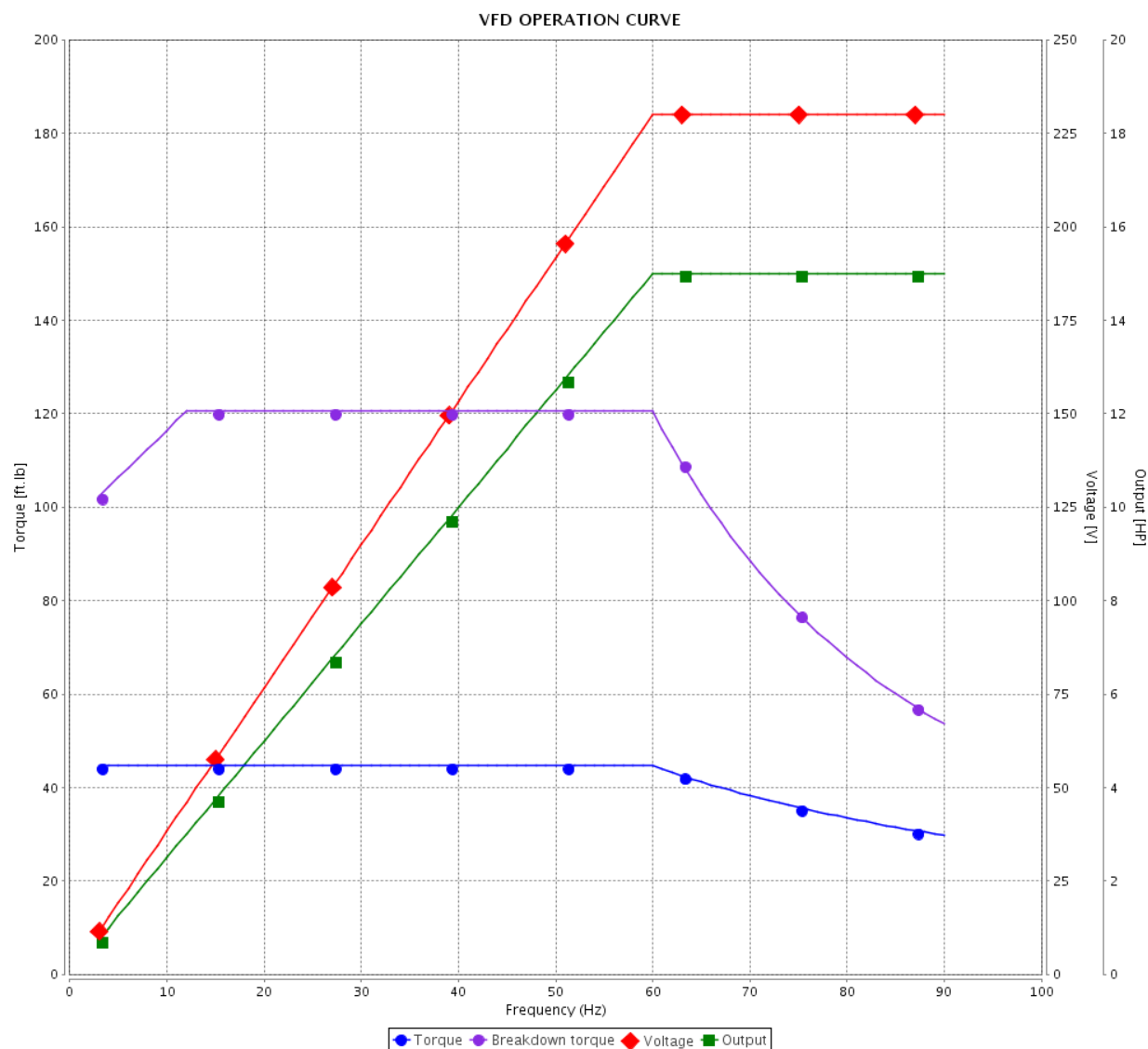


Customer :

Product line : W22 NEMA Premium Efficiency
Three-Phase

Product code : 12446945

Catalog # : 01518ET3E254TC-W22



Performance : 208-230/460 V 60 Hz 4P

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Insulation class : F
Service factor : 1.25
Temperature rise : 80 K
Design : B

Rev.	Changes Summary	Performed	Checked	Date
Performed by		Page 6 / 6 Revision		
Checked by				
Date	26/02/2020			

MODEL 01518ET3E254TC-W22

MADE IN MEXICO

12446945



Inverter Duty Motor
Severe Duty



FOR SAFE AREA
MOD.TE1BFOXON



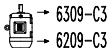
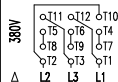
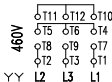
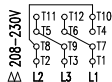
For 60Hz: Class I, Div 2, Gr. A, B, C and D - T3

Class I, Zone 2, IIC - T3

Class II, Div 2, Gr. F and G - T4

For 60Hz use on PWM, Gr. A, B, C, D and F,
VT 1000:1, CT 20:1, 1.0SF, T3A

PH 3	FR 254/6TC	HP(kW) 15.0(11.0)	Hz 60
V 230/460	A 36.0/18.0	RPM 1765	DES B
NEMA NOM EFF	92.4 %	INS. CL. F Δ T 80 K	IP55
ENCL TEFC	DUTY CONT.	AMB. 40°C	
CODE G	PF 0.83	SF 1.25	SFA 45.0/22.5
USABLE @208V	39.8 A	SF 1.15	SFA 45.8
15HP 11kW 50Hz 380V 21.5A 1455RPM SF1.00 EFF 90.2% (IE2)			



MOBIL POLYREX EM
13 g 20000 h

ALT 1000 m.a.s.l. 315 Lbs

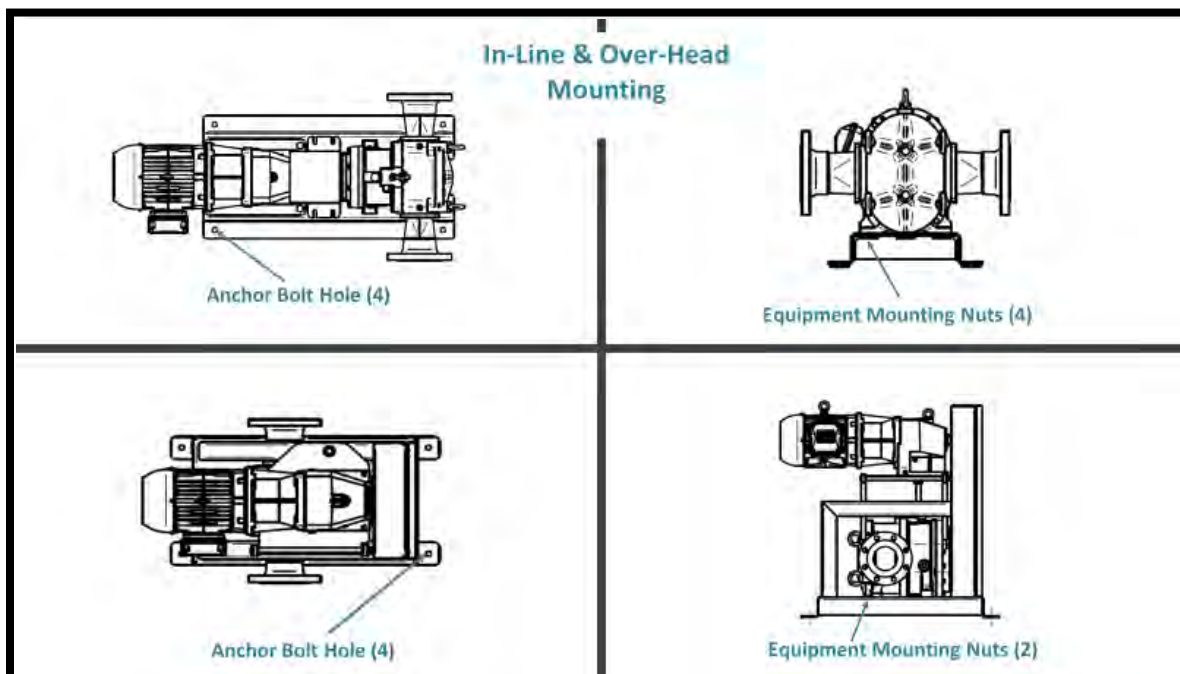
Installation Details

Installation Information

Boerger equipment base frames are designed to be installed without the use of grout, and grouting is not recommended. Leveling nuts beneath the pump frame are not permitted. In cases where leveling is necessary, or where there are gaps between the pump frame and base substrate, solid metal shims or spacers must be used, and anchors must be in full tension for rigid anchoring. Anchor bolt example below for reference.

Install the base frame onto the reinforced surface without subjecting it to stress, e.g. using four suitable anchor bolts and appropriate resin capsules or four other safe fixing systems suitable for the surface and the application.

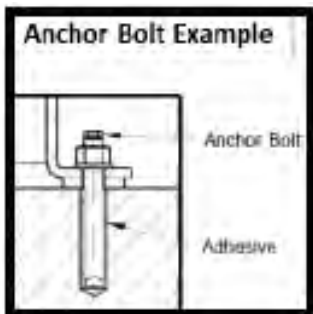
The equipment mounting nuts underneath the base frame must be accessible with a wrench from the front and back. If realignment is required or the machine is reinstalled on the base frame (e.g. following repairs or replacement), then it must be possible to hold the nuts in place with a wrench.



**Each frame has four (4) anchor bolt holes.
The holes sizes are as follows by Pump Series:**

- AL Pump Series Frame - .57" Ø
- PL Series Pump Frame - .6875 Ø
- * CL/EL/FL Pump Frame - .709 Ø

** EL Series pumps utilizing SK82 Reducer or larger will carry six (6) anchor holes. Please consult factory for further details.*



Spare Parts Supplied



Spare Parts & Special Tools Provided

Spare Parts:

- A. One (1) set of mechanical seals and o-rings for each pump model
- B. One (1) set of rotor tips and o-rings for each pump model
- C. One (1) set of axial protection plates for each pump model
- D. One (1) set of radial liners for each pump model
- E. One (1) set of special tools for each pump model

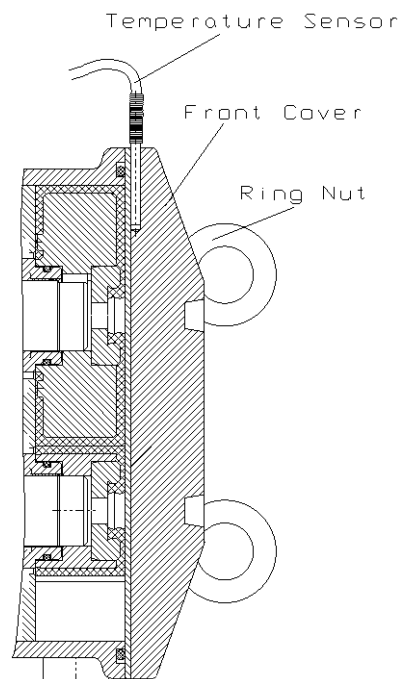
Dry Run Protection

Boerger dry run protection consists of a temperature sensing probe fitted into the front pump cover, and a digital temperature controller.

The controller is provided loose and is designed for flush panel mount.

Temperature Sensor

- Pyromation model RT01 RTD Assembly
- Grade B tolerance, 100 ohm resistance
- 316ss sheath, 6 mm diameter, 3" long
- 3-wire connection with 2" long stripped leads
- Teflon insulation, 84" long wire



Digital Temperature Controller (Provided Loose)

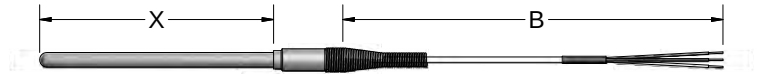
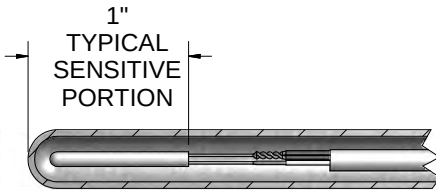
- Red Lion PXU PID Series
- Limit Controller
- 1/16 DIN Cutout
- Power 100-240 VAC
- LCD Display
- Relay outputs: One (1) normally-open & one (1) normally closed relay
- Analog output 4-20-mA
- Option: 1 alarm relay



RTD

Configuration Code RT01
RTD Assemblies with Extension Leadwire
 Configuration Code RT02
RTD Assemblies with Sheath Terminations

The RTD elements illustrated and described on this page are designed to measure temperature in a variety of process and laboratory applications. These RTDs are specifically designed for use in two different process temperature ranges and will provide accurate and repeatable temperature measurement through a broad range. Low range RTDs are constructed using Teflon®-insulated, silver-plated copper internal leads with potting compounds to resist moisture penetration. High range RTDs are constructed with nickel internal leads inside swaged MgO insulated cable to allow higher temperature measurements at the RTD element and provide higher temperature lead protection along the sheath. The following tables allow customer selection of standard element materials, tolerances, sheath diameters, mounting fittings and terminations. Custom-built assemblies with non-standard specifications are available upon request.



ORDER CODES

Example Order Number:

1-1 1-2(A) 1-3 1-4 00 13 T3084 2 (Z315)
R1T185L Z8 3 - 003 - Page RTD-2 - Page RTD-3 - Page RTD-4 - Page RTD-5

1-1 Single Platinum RTD Elements

CODE	TOLERANCE ^[1]	BASE RESISTANCE @ 0 °C (R ₀)	TEMPERATURE COEFFICIENT	CODE	6 mm O.D. Z8	(Z315)
LOW RANGE WIRE WOUND (-200 to 200) °C [-328 to 392] °F				1/8" O.D.	3/16" O.D.	1/4" O.D.
R1T185L	Grade B	100 Ω	α = 0.003 85 °C ⁻¹	28	38	48
R3T185L	Class AA	100 Ω	α = 0.003 85 °C ⁻¹	28	38	48
R5T185L	(1/5) Class B	100 Ω	α = 0.003 85 °C ⁻¹	28	38	48
R1T192L	Grade B	100 Ω	α = 0.003 92 °C ⁻¹	28	38	48
R3T192L	Class AA	100 Ω	α = 0.003 92 °C ⁻¹	28	38	48
LOW RANGE THIN FILM (-50 to 200) °C [-58 to 392] °F						
RBF185L	Class B	100 Ω	α = 0.003 85 °C ⁻¹	28	38	48
RAF185L	Class A	100 Ω	α = 0.003 85 °C ⁻¹	28	38	48
RBF195L	Class B	1000 Ω	α = 0.003 85 °C ⁻¹	28	38	48
HIGH RANGE WIRE WOUND (-200 to 600) °C [-328 to 1112] °F						
R1T185H	Grade B	100 Ω	α = 0.003 85 °C ⁻¹	28	38	48
RAT185H	Class A	100 Ω	α = 0.003 85 °C ⁻¹	28	38	48
R1T192H	Grade B	100 Ω	α = 0.003 92 °C ⁻¹	28	38	48

[1] Refer to RTD tolerance information in the general information section for calculations to determine specific tolerance at temperature.

1-1 Duplex Platinum RTD Elements

CODE	TOLERANCE ^[1]	BASE RESISTANCE @ 0 °C (R ₀)	TEMPERATURE COEFFICIENT	CODE	3/16" O.D.	1/4" O.D.	3/8" O.D.
LOW RANGE WIRE WOUND (-200 to 200) °C [-328 to 392] °F							
R1T285L	Grade B	100 Ω	α = 0.003 85 °C ⁻¹	38	48	68	
R3T285L	Class AA	100 Ω	α = 0.003 85 °C ⁻¹	38	48	68	
R5T285L	(1/5) Class B	100 Ω	α = 0.003 85 °C ⁻¹	38	48	68	
R1T292L	Grade B	100 Ω	α = 0.003 92 °C ⁻¹	38	48	68	
R3T292L	Class AA	100 Ω	α = 0.003 92 °C ⁻¹	38	48	68	
LOW RANGE THIN FILM (-50 to 200) °C [-58 to 392] °F							
RBF285L	Class B	100 Ω	α = 0.003 85 °C ⁻¹	38	48	68	
RAF285L	Class A	100 Ω	α = 0.003 85 °C ⁻¹	38	48	68	
RBF295L	Class B	1000 Ω	α = 0.003 85 °C ⁻¹	38	48	68	
HIGH RANGE WIRE WOUND (-200 to 600) °C [-328 to 1112] °F							
R1T285H	Class B	100 Ω	α = 0.003 85 °C ⁻¹	38	48	68	
RAT285H	Class A	100 Ω	α = 0.003 85 °C ⁻¹	38	48	68	
R1T292H	Grade B	100 Ω	α = 0.003 92 °C ⁻¹	38	48	68	

[1] Refer to RTD tolerance information in the general information section for calculations to determine specific tolerance at temperature.

1-2 Available Sheath Diameters 316SS

1-4 Length

CODE
3 Digit 'X' Length

1-3 Element Connection

CODE	DESCRIPTION
2	2-wire
3	3-wire
4 ^[1]	4-wire

[1] Not available in duplex

1-2A

CODE	NOMINAL SHEATH DIAMETER (inches)	TIP DIA. O.D. (inches)	TIP LENGTH (inches)
88R48	1/2	1/4	1 1/4
68R38	3/8	3/16	1 1/4
48R28	1/4	1/8	1 1/4

REDUCED-TIP RTD's

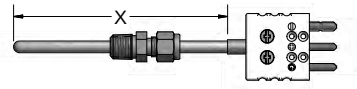
Table 1-2A lists RTD elements with reduced tip sheaths. To order, use order code numbers from Tbl. 1-2A in place of straight sheath order code numbers from Tbl. 1-2. Other reduced tips are available upon request. EXAMPLE: R1T185L88R483-006.

Teflon® is a registered trademark of E. I. du Pont de Nemours and Company.

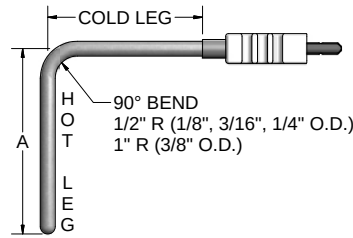
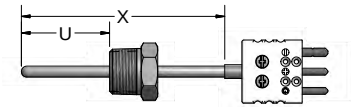


Select Sheath Mounting or Bend Options as desired from tables below.

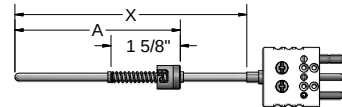
COMPRESSION FITTING



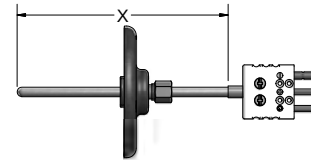
FIXED BUSHING



BAYONET CAP and SPRING (OPTION 13A)



ADJUSTABLE FLANGE (OPTION 14)



ORDER CODES

Example Order Number:

R5T185L483-006 -

2
01A,304

PAGE
RTD 3

PAGE
RTD 4

PAGE
RTD 5

2-1 No Fitting or Bend Options

CODE	00
------	----

2-2 One-time Adjustable Compression Fittings

CODE	TYPE	NPT SIZE (inches)	PRESSURE RATED	AVAILABLE SHEATH DIAMETERS (inches)
01A	303 stainless steel	1/8	NO	1/8, 3/16, 1/4
05A	316 stainless steel	1/8	YES	1/8, 3/16, 1/4
05B	316 stainless steel	1/4	YES	1/8, 3/16, 1/4, 3/8
05C	316 stainless steel	1/2	YES	1/8, 1/4, 3/8
15A	Brass	1/8	NO	1/8, 3/16, 1/4
15B	Brass	1/4	NO	3/16, 1/4, 3/8
15C	Brass	1/2	NO	1/4, 3/8

2-3 Re-adjustable Compression Fittings

CODE	TYPE	NPT SIZE (inches)	AVAILABLE SHEATH DIAMETERS (inches)
10A	303 stainless steel	1/8	1/8, 3/16
10B	303 stainless steel	1/4	1/4, 3/8
10C	303 stainless steel	1/2	1/4, 3/8
12A	316 stainless steel	1/8	1/8, 3/16, 1/4
12B	316 stainless steel	1/4	1/8, 3/16, 1/4, 3/8
12C	316 stainless steel	1/2	1/8, 1/4, 3/8
11A	Brass	1/8	1/8, 3/16, 1/4
11B	Brass	1/4	1/8, 3/16, 1/4, 3/8
11C	Brass	1/2	1/4, 3/8
19C	Spring-loaded SS well fitting	1/2	3/16, 1/4

Teflon® gland standard 204 °C [400 °F] max. For lava gland 649 °C [1200 °F] max. opt. 10A and 10B only use letter suffix "L" after compression fitting order code. EXAMPLE: 10AL for lava gland.

2-6 Miscellaneous Options

CODE	TYPE	AVAILABLE SHEATH DIAMETER (inches)
13A __ [1]	Spring-loaded bayonet fitting	1/8, 3/16
14	Adjustable flange with brass compression fitting	1/8, 3/16, 1/4, 3/8
16A	Spring-loaded adjustable bayonet compression fitting	1/8

[1] When ordering fixed bayonet fitting specify dimension "A". EXAMPLE: order code 13A06 is for a fixed bayonet adapter with 6" A Dimension.

2-5 Fixed Bushings

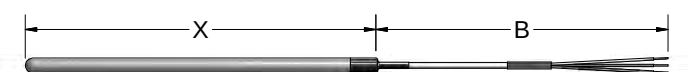
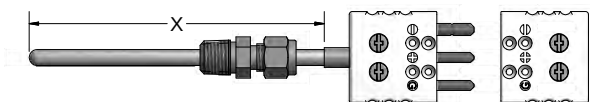
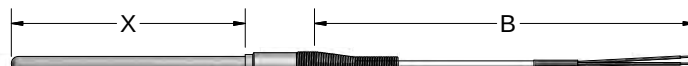
CODE	MOUNTING THREAD NPT (inches)	AVAILABLE SHEATH DIAMETERS (inches)
316 SS		
8A __ [1]	1/8	1/8, 3/16, 1/4
8B __ [1]	1/4	1/8, 3/16, 1/4, 3/8
8C __ [1]	1/2	1/8, 3/16, 1/4, 3/8
8D __ [1]	3/4	1/8, 3/16, 1/4, 3/8

[1] When ordering fixed bushings, specify order code above, plus insertion length "U", as measured from hot tip to bottom of threaded bushing. EXAMPLE: order code 8A06 is 1/8" NPT, 316 SS bushing located 6" from hot tip.

2-4 Sheath Bends

CODE	DESCRIPTION
2 __	Sheath bent 45°
3 __	Sheath bent 90°
2" minimum hot leg length	
When ordering bend options, specify hot leg dim. "A". EXAMPLE: order code 206 is a 45° bend with 6" hot leg. Total sheath length is Table 1 "X" length = hot leg plus cold leg.	

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RT02

ORDER CODES

RT01

Example Order Number:

R5T185L483-006-00 - ³⁻¹4, MC or R5T185L483-006-01A,304 - ³⁻²16 - PAGE RTD-4 - PAGE RTD-5

3-1 Plug and Jack Sheath Terminations

CODE	DESCRIPTION
4 ^[1]	Standard plug
5 ^[1]	Standard jack
6 ^[2]	Miniature plug
7 ^[2]	Miniature jack
Options	
MC	Mating connector
CL	Compression L bracket to hold plug to sheath
[1] If used with 3/8" O.D., option CL must be specified	
[2] Not available with 1/4" O.D. or 3/8" O.D. sheath	

3-1 Sheath Terminations

CODE	DESCRIPTION
22 ^[1]	3" individual leads with terminal pins
[1] High temp RTDs are supplied with 1" long transition	

3-2 Leadwire transitions

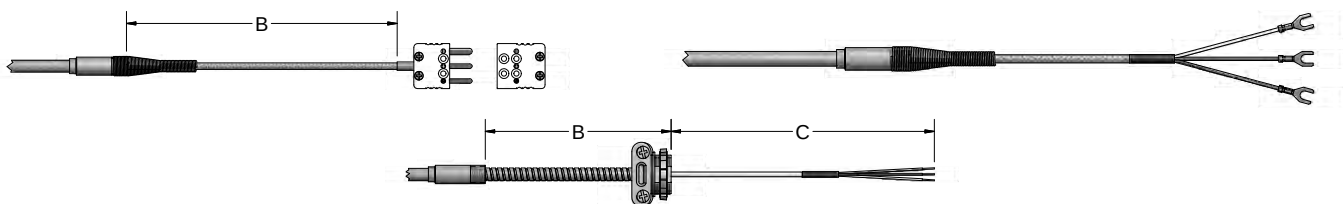
(Requires Table 4 and 5 selections)

CODE	DESCRIPTION
13 ^[1]	Same size transition with heat-shrink tubing 104 °C [220 °F]
15	Extension leadwire transition with relief spring 204 °C [400 °F]
16	Extension leadwire transition with heat-shrink tubing 104 °C [220 °F]
18 ^[1]	Same size transition without heat-shrink tubing 204 °C [400 °F]
19	Extension leadwire transition without spring or heat-shrink tubing 204 °C [400 °F]
Options	
HT ^[2]	High temperature potting 538 °C [1000 °F] not available with option 13 or 16
[1] Not available with flex armor	
[2] Not available with option 13 or 16. When specifying high temp potting with Flex Armor option 19 must be selected.	

3-2 Threaded Fittings with Extension Leadwire (Requires Table 4 and 5 selections)

CODE	DESCRIPTION
6HN23	1/2" x 1/2" NPT steel hex nipple
8HN23	1/2" x 1/2" NPT stainless steel hex nipple
9HP23	1/2" NPT stainless steel bushing (no process threads)
8RNDC23	3/4" process x 1/2" NPT stainless steel hex nipple

Select desired leadwire type by order code number, followed by desired length in inches.



ORDER CODES

Example Order Number:

R5T185L483-006-01A,304-16

T3

4

036

-

5

PAGE
RTD-5

4 Extension Leadwire Type and B + C Dimension

CODE	DESCRIPTION	TEMP. RATING
FIBERGLASS		
F3J_ _ _	Fiberglass insulation - individual leads - stranded conductor (12" limit)	482 °C [900 °F]
F3_ _ _	Fiberglass insulation - stranded conductor	
F3A_ _ _	Fiberglass insulation - stranded conductor - flexible armor	
F3B_ _ _	Fiberglass insulation - stranded conductor - stainless steel overbraid	
TEFLON®		
T3J_ _ _	Teflon® insulation - individual leads - stranded conductor (12" limit)	204 °C [400 °F]
T3_ _ _	Teflon® insulation - stranded conductor	
T3A_ _ _	Teflon® insulation - stranded conductor - flexible armor	
T3B_ _ _	Teflon® insulation - stranded conductor - stainless steel overbraid	
M3_ _ _	Teflon® insulation - stranded conductor - stainless steel overbraid - Teflon® insulation	
T3M_ _ _	Teflon® insulation - stranded conductor - mylar shield	
T3MA_ _ _	Teflon® insulation - stranded conductor - mylar shield - flexible armor	
KAPTON®		
K3_ _ _	Kapton® insulation - stranded conductor	316 °C [600 °F]
K3A_ _ _	Kapton® insulation - stranded conductor - flexible armor	
K3B_ _ _	Kapton® insulation - stranded conductor - stainless steel overbraid	
SILICON RUBBER		
S3_ _ _	Teflon® insulation - stranded conductor - silicon rubber	204 °C [400 °F]
COIL CORDS		
C3060	PVC insulation - stranded conductor - coil cord - 60" extended length	104 °C [220 °F]
C3120	PVC insulation - stranded conductor - coil cord - 120" extended length	

084 = 84" wire length (up to 250' of additional wire may be field installed - 22 ga.)

Insert wire code number and 3 digit 'B' length in inches EXAMPLE: T3036 = 36" B length

For assemblies requiring leadwire beyond the flexible armor (illustrated in 'C' in drawing), insert 3 digit 'C' length after armor length.
EXAMPLE: F3A036-012 = 36" B length with additional 12" 'C' length leads beyond armor.

All insulated leadwires in flexible armor are available with either extruded PVC or Teflon® covering over the flexible armor.
Substitute suffix codes T (Teflon®) or P (PVC) for the suffix 'A' code above. EXAMPLE: T3T is Teflon® covered armor.

Teflon® and Kapton® are registered trademarks of E. I. du Pont de Nemours and Company.



Select desired leadwire termination and options (if desired), by order code numbers below.

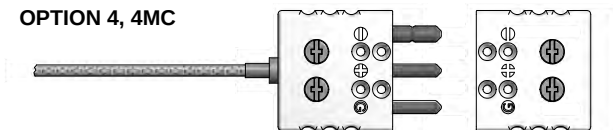
OPTION 3



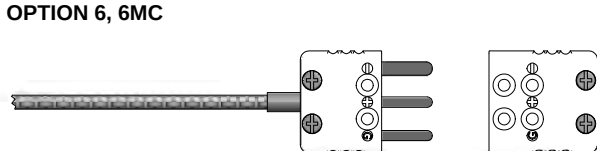
OPTION 8



OPTION 4, 4MC



OPTION 6, 6MC



ORDER CODES

Example Order Number:

R5T185L483-006-01A,304-16-T3036 -

5-1 5-2
4, MC

5-1 Terminations

CODE	DESCRIPTION
0	Leads not stripped
2	2" split leads, 1/4" stripped
3	2" split leads with spade lugs
4	Standard plug
5	Standard jack
6	Miniature plug
7	Miniature jack
8	2" split leads with 1/4" female quick disconnects

5-2 Options

CODE	DESCRIPTION
BX	1/2" NPT BX connector with Options 0, 2, 3, or 8
CC	Plug or jack secured to leads with cable clamp
CG	Cord grip (1/2" NPT PVC)
MC	Mating connector
RB	Rubber boot

(Z315) 6mm O.D.

PXU PID Controller

Red Lion Automation Series



▶▶▶ Temperature/Process Controllers

Red Lion's versatile PXU Series controllers are ideal for a variety of industrial applications - from simple on/off to full PID control and more.

Our PXU PID controllers offer many unique features including universal inputs 1/16, 1/8 and 1/4 DIN models, as well as one of the largest dual line displays in the industry. Other features include on-demand auto tuning, two control outputs for reverse or direct application and easy setup with Crimson 2.1 software. Options include ramp/soak capability, enabling users to change and hold temperature and process variables for specific time periods, easily integrating into time-stepped processes. Other options include a second analog input, second analog output as well as a current transducer (CT) input.

As one of the most complete PID controller lines in the industry, the PXU Series is an ideal choice for machine builders and process engineers alike.



APPLICATIONS

- > Food & Beverage
- > Factory Automation
- > Manufacturing
- > Plastics
- > Automotive

PRODUCT HIGHLIGHTS

- > Ramp/Soak Capability
- > 2nd Analog Input
- > 2nd Analog Output
- > Current Transducer (CT) Input
- > Full PID Control
- > Dual Relay Alarms
- > UL Listed, 61010-1 Approved

FEATURES & BENEFITS

- > Universal Inputs
 - Thermocouple
 - RTD
 - 0-10 VDC
 - 4 – 20 mA (0 – 50 mA)
- > Full PID Control
 - On demand auto-tuning
 - Independently configured outputs
- > Easy Programming
 - Front panel push-button or Crimson software
- > Digital Pot Functionality
- > AC or DC Powered Models
 - 100 to 240 VAC
 - 24 VDC
- > Reduced Case Depth
- > IP 65 Ingress Protection

industrial
automation



▶▶▶ PXU PID Controller Specifications

DISPLAY PROPERTIES

LCD negative image transmissive with backlighting. Top (process) display with orange backlighting, bottom (parameter) display with green backlighting.

4 digits each line

1/4 DIN Model Digit Size: Line 1 - 0.87" (22 mm);
Line 2 - 0.55" (14 mm)

1/8 DIN Model Digit Size: Line 1 - 0.47" (12 mm);
Line 2 - 0.47" (12 mm)

1/16 DIN Model Digit Size: Line 1 - 0.43" (11 mm);
Line 2 - 0.27" (7.0 mm)

POWER CONNECTION

AC Power: 100 to 240 VAC -20/+8 %, 50/60 Hz, 5 VA
DC Power: 24 VDC, $\pm 10\%$, 5 W

INPUT CAPABILITIES

Temperature/RTD Indication Accuracy:
 $\pm (0.3\% \text{ of span, } +1^\circ\text{C}) \text{ at } 25^\circ\text{C}$

Thermocouple Inputs:

Types: T, E, J, K, R, S, B, N, L, U, and TXK

Resolution: 1° for types R, S, B and 1° or 0.1° for all other types

RTD Inputs:

Type: 2 or 3 wire

Excitation: 180 μ A

Process Inputs:

0-5 or 0-10 VDC

0-20 or 4-20 mA

0-50 mV

CT Input (Optional): CT is included with this option

Type: Single phase, full wave monitoring of load currents

Input: 0 to 25 mA AC

Display Scale Range: 1.0 to 999.9 amperes

CT Rating: Current Ratio: 40 A/30.7 mA AC Turn Ratio: 1:1300

Remote Input

Selectable: 0-5 V, 1-5 V, 0-10 V, 0-20 mA, 4-20 mA

A/D Conversion Rate: 10 samples per second

COMMUNICATION

RS-485 serial communication (optional)

OUTPUTS

Time proportioning or DC analog (mA and mV)

Alarms: 2 relay alarm outputs.

Type: Form A or Form C, model and alarm dependent

Contact Rating: 3 A @ 250 VAC

CERTIFICATIONS & COMPLIANCES

CE Approved

EN 61326-1 immunity to industrial locations

Emission CISPR 11 Class A

EN 61010-1

RoHS compliant

UL Listed: File #E179259

IP65 enclosure rating (face only)

ENVIRONMENTAL

Operating Temperature Range: 0 to 50°C

Storage Temperature Range: -20 to 65°C

Operating and Storage Humidity: 80% max relative humidity (non-condensing) from 0°C to 50°C

Vibration Resistance: Operational 10 to 55 Hz, 1 g

Shock Resistance: Operational 30 g

Altitude: Up to 2000 meters

MECHANICAL

Construction: Black plastic alloy case and panel latch. Black plastic textured bezel with transparent display window. Controller meets IP65 requirements for indoor use when properly installed. Installation Category II, Pollution Degree 2.

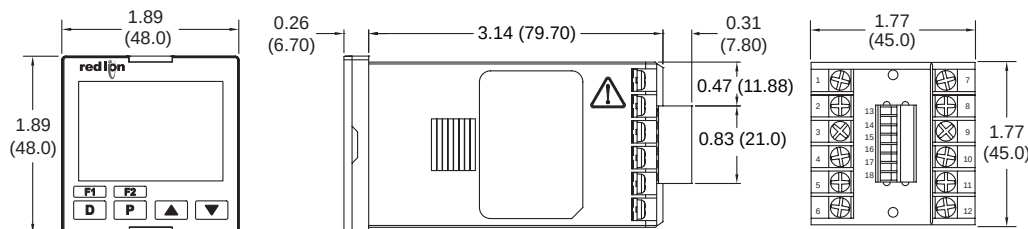
Weight: 1/4 DIN: 11.0 oz (312 g)

1/8 DIN: 7.8 oz (221 g)

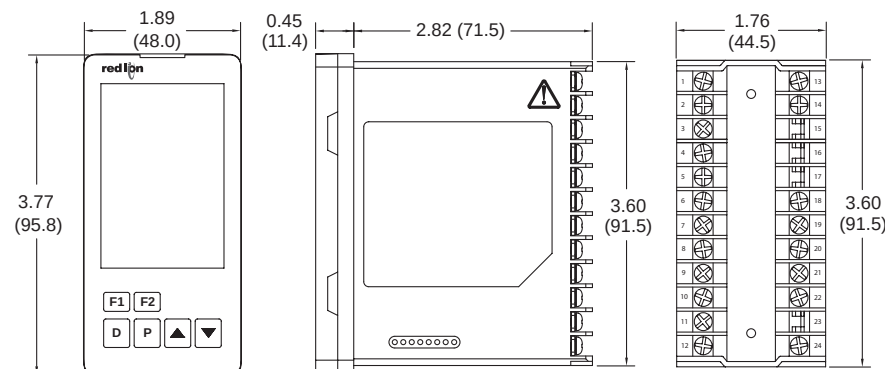
1/16 DIN: 5.3 oz (150 g)

DIMENSIONS In inches (mm)

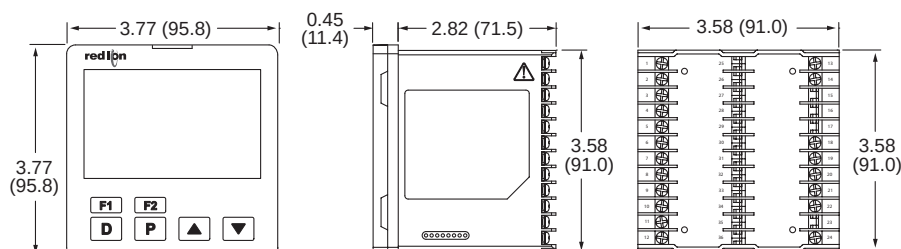
1/16 DIN



1/8 DIN



1/4 DIN



▶▶▶ PXU PID Controller Specifications

ORDERING GUIDE

DIN SIZE	MAIN CONTROL OUTPUT 1	SECONDARY CONTROL OUTPUT 2	USER INPUT(S)	REMOTE SETPOINT INPUT	CT INPUT	ANALOG OUTPUT (RETRANS)	RS 485	PART NUMBERS	
								100 to 240 VAC	24 VDC
1/16 DIN	Relay							PXU10020	PXU100B0
	Relay	Relay	2				Yes	PXU11A20	PXU11AB0
	Relay	Relay	1		Yes		Yes	PXU11B20	PXU11BB0
	Relay	Relay	1			Yes	Yes	PXU11C20	PXU11CB0
	Relay	Relay	1	Yes			Yes	PXU11D20	PXU11DB0
	Logic/SSR							PXU20020	PXU200B0
	Logic/SSR	Relay	2				Yes	PXU21A20	PXU21AB0
	Logic/SSR	Relay	1		Yes		Yes	PXU21B20	PXU21BB0
	Logic/SSR	Relay	1			Yes	Yes	PXU21C20	PXU21CB0
	Logic/SSR	Relay	1	Yes			Yes	PXU21D20	PXU21DB0
	4-20 mA							PXU30020	PXU300B0
	4-20 mA	Relay	2				Yes	PXU31A20	PXU31AB0
	4-20 mA	Relay	1		Yes		Yes	PXU31B20	PXU31BB0
	4-20 mA	Relay	1			Yes	Yes	PXU31C20	PXU31CB0
	4-20 mA	Relay	1	Yes			Yes	PXU31D20	PXU31DB0
	0-10 VDC							PXU40020	PXU400B0
	0-10 VDC	Relay	2				Yes	PXU41A20	PXU41AB0
	0-10 VDC	Relay	1		Yes		Yes	PXU41B20	PXU41BB0
	0-10 VDC	Relay	1			Yes	Yes	PXU41C20	PXU41CB0
	0-10 VDC	Relay	1	Yes			Yes	PXU41D20	PXU41DB0
1/4 DIN	Relay	Relay	2				Yes	PXU11A50	PXU11AE0
	Logic/SSR	Relay	1		Yes		Yes		PXU21BE0
	Logic/SSR	Relay	1			Yes	Yes		PXU21CE0
	Logic/SSR	Relay	1	Yes			Yes		PXU21DE0
	4-20 mA	Relay	2				Yes	PXU31A50	PXU31AE0
	4-20 mA	Relay	1		Yes		Yes		PXU31BE0
	4-20 mA	Relay	1			Yes	Yes		PXU31CE0
	4-20 mA	Relay	1	Yes			Yes		PXU31DE0
	0-10 VDC	Relay	2				Yes	PXU41A50	PXU41AE0
	0-10 VDC	Relay	1		Yes		Yes		PXU41BE0
	0-10 VDC	Relay	1			Yes	Yes		PXU41CE0
	0-10 VDC	Relay	1	Yes			Yes		PXU41DE0
1/8 DIN	Relay							PXU10030	PXU100C0
	Relay	Relay	2				Yes	PXU11A30	PXU11AC0
	Relay	Relay	1		Yes		Yes	PXU11B30	PXU11BC0
	Relay	Relay	1			Yes	Yes	PXU11C30	PXU11CC0
	Relay	Relay	1	Yes			Yes	PXU11D30	PXU11DC0
	Logic/SSR							PXU20030	PXU200C0
	Logic/SSR	Relay	2				Yes	PXU21A30	PXU21AC0
	Logic/SSR	Relay	1		Yes		Yes	PXU21B30	PXU21BC0
	Logic/SSR	Relay	1			Yes	Yes	PXU21C30	PXU21CC0
	Logic/SSR	Relay	1	Yes			Yes	PXU21D30	PXU21DC0
	4-20 mA							PXU30030	
	4-20 mA	Relay	2				Yes	PXU31A30	PXU31AC0
	4-20 mA	Relay	1		Yes		Yes	PXU31B30	PXU31BC0
	4-20 mA	Relay	1			Yes	Yes	PXU31C30	PXU31CC0
	4-20 mA	Relay	1	Yes			Yes	PXU31D30	PXU31DC0
	0-10 VDC							PXU40030	PXU400C0
	0-10 VDC	Relay	1		Yes		Yes	PXU41B30	PXU41BC0
	0-10 VDC	Relay	1			Yes	Yes	PXU41C30	PXU41CC0
	0-10 VDC	Relay	1	Yes			Yes	PXU41D30	PXU41DC0

Only stocked part numbers are listed. Contact factory for availability of non-stock models. In order to program the unit using Crimson, the unit must be purchased with the RS 485 option.

ORDERING GUIDE

DIN SIZE	MAIN CONTROL OUTPUT 1	SECONDARY CONTROL OUTPUT 2	USER INPUT(S)	REMOTE SETPOINT INPUT	CT INPUT	ANALOG OUTPUT (RETRANS)	RS 485	PART NUMBERS	
								100 to 240 VAC	24 VDC
1/16 DIN	Relay							PXU10020	PXU100B0
	Relay	Relay	2				Yes	PXU11A20	PXU11AB0
	Relay	Relay	1		Yes		Yes	PXU11B20	PXU11BB0
	Relay	Relay	1			Yes	Yes	PXU11C20	PXU11CB0
	Relay	Relay	1	Yes			Yes	PXU11D20	PXU11DB0
	Logic/SSR							PXU20020	PXU200B0
	Logic/SSR	Relay	2				Yes	PXU21A20	PXU21AB0
	Logic/SSR	Relay	1		Yes		Yes	PXU21B20	PXU21BB0
	Logic/SSR	Relay	1			Yes	Yes	PXU21C20	PXU21CB0
	Logic/SSR	Relay	1	Yes			Yes	PXU21D20	PXU21DB0
	4-20 mA							PXU30020	PXU300B0
	4-20 mA	Relay	2				Yes	PXU31A20	PXU31AB0
	4-20 mA	Relay	1		Yes		Yes	PXU31B20	PXU31BB0
	4-20 mA	Relay	1			Yes	Yes	PXU31C20	PXU31CB0
	4-20 mA	Relay	1	Yes			Yes	PXU31D20	PXU31DB0
	0-10 VDC							PXU40020	PXU400B0
	0-10 VDC	Relay	2				Yes	PXU41A20	PXU41AB0
	0-10 VDC	Relay	1		Yes		Yes	PXU41B20	PXU41BB0
	0-10 VDC	Relay	1			Yes	Yes	PXU41C20	PXU41CB0
	0-10 VDC	Relay	1	Yes			Yes	PXU41D20	PXU41DB0
1/4 DIN	Relay	Relay	2				Yes	PXU11A50	PXU11AE0
	Logic/SSR	Relay	1		Yes		Yes		PXU21BE0
	Logic/SSR	Relay	1			Yes	Yes		PXU21CE0
	Logic/SSR	Relay	1	Yes			Yes		PXU21DE0
	4-20 mA	Relay	2				Yes	PXU31A50	PXU31AE0
	4-20 mA	Relay	1		Yes		Yes		PXU31BE0
	4-20 mA	Relay	1			Yes	Yes		PXU31CE0
	4-20 mA	Relay	1	Yes			Yes		PXU31DE0
	0-10 VDC	Relay	2				Yes	PXU41A50	PXU41AE0
	0-10 VDC	Relay	1		Yes		Yes		PXU41BE0
	0-10 VDC	Relay	1			Yes	Yes		PXU41CE0
	0-10 VDC	Relay	1	Yes			Yes		PXU41DE0
1/8 DIN	Relay							PXU10030	PXU100C0
	Relay	Relay	2				Yes	PXU11A30	PXU11AC0
	Relay	Relay	1		Yes		Yes	PXU11B30	PXU11BC0
	Relay	Relay	1			Yes	Yes	PXU11C30	PXU11CC0
	Relay	Relay	1	Yes			Yes	PXU11D30	PXU11DC0
	Logic/SSR							PXU20030	PXU200C0
	Logic/SSR	Relay	2				Yes	PXU21A30	PXU21AC0
	Logic/SSR	Relay	1		Yes		Yes	PXU21B30	PXU21BC0
	Logic/SSR	Relay	1			Yes	Yes	PXU21C30	PXU21CC0
	Logic/SSR	Relay	1	Yes			Yes	PXU21D30	PXU21DC0
	4-20 mA							PXU30030	
	4-20 mA	Relay	2				Yes	PXU31A30	PXU31AC0
	4-20 mA	Relay	1		Yes		Yes	PXU31B30	PXU31BC0
	4-20 mA	Relay	1			Yes	Yes	PXU31C30	PXU31CC0
	4-20 mA	Relay	1	Yes			Yes	PXU31D30	PXU31DC0
	0-10 VDC							PXU40030	PXU400C0
	0-10 VDC	Relay	1		Yes		Yes	PXU41B30	PXU41BC0
	0-10 VDC	Relay	1			Yes	Yes	PXU41C30	PXU41CC0
	0-10 VDC	Relay	1	Yes			Yes	PXU41D30	PXU41DC0

Only stocked part numbers are listed. Contact factory for availability of non-stock models. In order to program the unit using Crimson, the unit must be purchased with the RS 485 option.

Factory Testing

Factory Testing

Each pump, will receive the following factory testing:

Performance Test
ANSI/HI 3.6-2022
Type IV, Level A
With Certificate

Vibration Test
DIN ISO 20816-1
In Combo w/ Performance Test
w/ Certificate

Noise Test dB (A)
DIN EN 61672-1/API 682
In Combo w/ Performance Test
w/ Certificate

The testing certificates will be made available after submittals/equipment has been approved and the manufacturing is completed on all pumps allowing for testing.

Boerger Performance Test Procedure

1. Application

1.1. This test procedure shall apply to Boerger rotary lobe pumps

2. Scope

2.1. This test procedure is based on the American National Standard for Rotary Pump Tests:
ANSI/HI 3.6-2022

3. Test setup

3.1. The test setup shall include provisions to vary the inlet and outlet pressures and to measure the following:

3.1.1. Liquid temperature at pump inlet

3.1.2. Pressure at pump inlet and outlet

3.1.3. Rate of flow (Types III and IV)

3.1.4. Pump speed

3.1.5. Pump input power (Type IV)

4. Test procedure

4.1. Prior to the test run, data shall be obtained, where applicable, according to Figure 3.6.5.11.2 of ANSI/HI 3.6-2022

4.2. Pre-test checks are carried out when the installation is complete and before test runs are begun. Checks are made for:

4.2.1. Proper pump mounting

4.2.2. Pump and piping alignment

4.2.3. Driver rotation

4.2.4. Freedom of obstruction and leaks in piping system

4.2.5. Determine that pump and driver are free turning

4.3. Test shall commence at one or more test point (Types I and II) or three or more test points (Types III and IV).

4.3.1. The outlet pressure shall be maintained constant, and the pump shall be operated at additional speed values, with rate of flow and power readings recorded for each.

4.3.2. Values are tabulated according to Table 3.6.5.7.3 of ANSI/HI 3.6-2022

5. Records

5.1. Type I

5.1.1. Test data is not furnished to the purchaser

5.2. Type II

5.2.1. A certificate of compliance shall be furnished

5.3. Type III

5.3.1. A performance curve shall be furnished

5.4. Type IV

5.4.1. A performance datasheet with record of data taken and a performance curve shall be furnished

Börger Guideline BG-VT

13th November 2018 - USA Version

Vibration Testing on Börger Pump-Systems

1. Introduction

Offline vibration monitoring is useful for pump systems to assess their running condition. A typical application is cyclic monitoring of pump and drive vibration as part of a maintenance plan. A maintenance interval may be one day or some days up to weeks, depending on type and importance of the pump-system. System condition monitoring at an early stage helps to prevent unexpected breakdowns. Thus expensive secondary damage and manufacturing loss can be avoided. Börger recommends high quality vibration meters with magnetic clamp sensors. Please follow the recommendations of this guideline. The guideline is made in accordance to the international standard ISO 10816.

ISO 10816: Mechanical vibration -- Evaluation of machine vibration by measurements on non-rotating parts

2. General

Before using a Vibration Meter, suitable measuring points at the machine need to be selected. For this purpose it is recommended to turn to specialists with experiences in the field of vibration monitoring on machinery. In general it is advisable to measure vibration near to its source to minimize distortion of the signal to be measured by transmitting mechanical components. Suitable measuring points are rigid components, for instance the housing of bearings or gearboxes. Especially at vibration monitoring of roller bearings the distance from the pickup to the bearing is to minimize, with only a small number of junctions. This way you can avoid distortion of the frequency characteristics of the measuring signal. Unsuitable for these measurements are fixing points at lightweight, flexible and soft components.

3. Measuring Points in accordance to ISO 10816-1

The standard ISO 10816-1 recommends for vibration measurements on machines the housing of bearings or nearby measuring points. For routine monitoring it is sufficient in many cases to measure vibration only in vertical or in horizontal direction. Rigid mounted machines with horizontal shafts have their highest vibration levels mostly in horizontal direction. Flexible mounted machines may have high vertical components of vibration, too. For inspections vibration should be measured in all three directions (vertical, horizontal and axial) at all bearings. For better explanation see Figures 1-3 on page 2.

3.1. Machine Vibration vs. Hydraulic Pulsation

Hydraulic pulsation is inherent in all rotary lobe pumps. In cases of high discharge pressure or other conditions, hydraulic pulsations may affect vibration readings (cannot discern one from the other). In such cases, the machine may be tested unloaded as provenance of actual machine vibration, with corresponding measurements recorded, and allowances for hydraulic pulsations noted for out-of-range readings with pump under load.

Recommended measuring points in accordance to ISO 10816-1 (Figure 1 – 3)

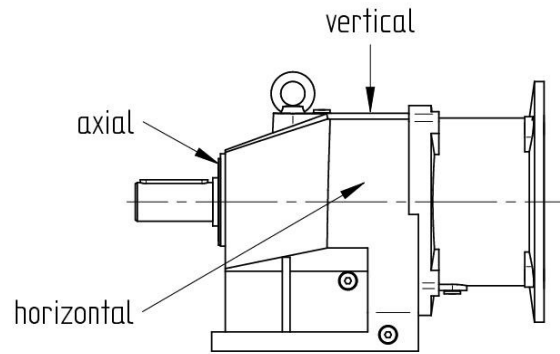


Figure 1: Measuring points on reduction gears

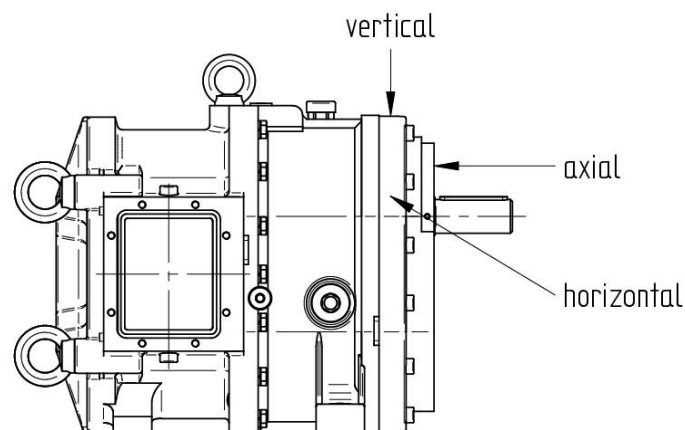


Figure 2: Measuring points on lobe pumps

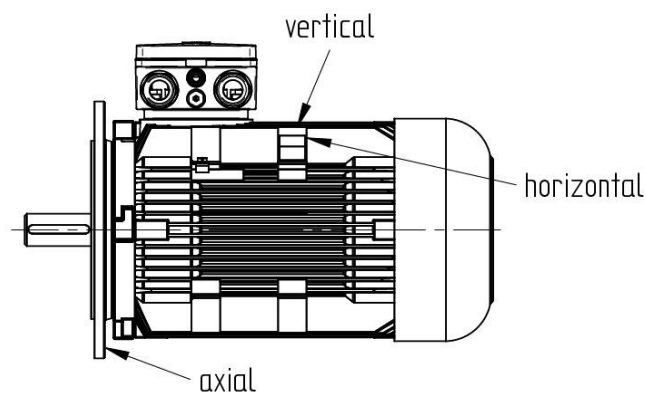


Figure 3: Measuring points on electric motors

4. Reference Values

If no reference values of vibration severity are available on the relevant machine, you may refer to the recommendations of ISO 10816-1 or API 676 (see below). Here you will find permissible values of the vibration severity of different machine types. The basis of the assessment is the maximum value of all measured points on the machine.

<u>4.1 Pumps – separate drive + rigid foundation (f.e. bolted)</u>	<u>Max. Cont. Ref. Val. [mm/s]</u>
P < 15 kW – new machine condition	<u>3,5</u>
P < 15 kW – unlimited long term operation allowed	4,5
P < 15 kW – short term operation allowed	7,1
P > 15 kW – new machine condition	<u>4,5</u>
P > 15 kW – unlimited long term operation allowed	7,1
P > 15 kW – short term operation allowed	11,0
<u>4.2 Pumps – separate drive + flex. foundation (f.e. damped)</u>	<u>Max. Cont. Ref. Val. [mm/s]</u>
P < 15 kW – new machine condition	4,5
P > 15 kW – new machine condition	7,1
<u>4.3 Pumps on Börger Test Bench – flex. foundation (clamped)</u>	<u>Max. Check Values [mm/s]</u>
P < 15 kW – new machine condition	<u>4,5</u>
P > 15 kW – new machine condition	<u>7,1</u>

5. Recommended way of machine monitoring

1. Check the installation
2. Estimate Reference Values [mm/s]

Reference Values above the Max. Continuous Values due to an improper installation.
In that case there must be a problem with:

- foundation
- clutch alignment
- external vibrations

3. Put the measured values into a vibration data sheet (see following example)
4. If a measured value gets higher than 125% of the reference value, a system check is recommended

6. Vibration Data Sheet (Example)

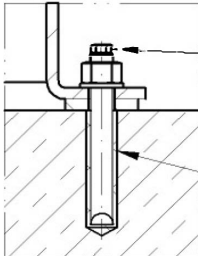
<u>System:</u>	Börger Lobe Pump PL 100 - SEW R67 geared drive (4,0 kW, 397 rpm)		
<u>Part:</u>	Polymer Pump – TAG-13-A2		
<u>Foundation:</u>	rigid		
<u>Max. Cont. Val.:</u>	4,5 [mm/s]	<u>Resp.:</u> H. Doelz	<u>Date:</u> 22.10.2013
<u>Direction</u>	<u>Check Value (+25%)</u>	<u>Measured [mm/s]</u>	<u>Sign of Operator</u>
Vertical:	5,6	1,9	#D
Horizontal:	5,6	2,0	#D
Axial:	5,6	2,5	#D

In general each pump, gear and motor has own values, depending on many different circumstances. For good maintenance practice it is important to estimate reference values and check values for each pump. If you rebuild your system you must estimate new values.

In accordance to this it is dangerous to operate with factory values. Factory values are only for quality control purposes. The values will not give to customers for maintenance practices.

Anchor Bolts

Pump Series: AL; AN

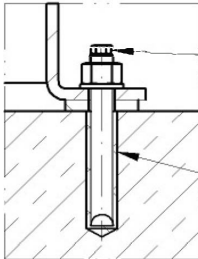


Anchor Bolts:
Hilti HAS-R 316 Stainless Steel
3/8" X 5-1/8"

Hilti HIT-HY 200 Adhesive

Leveling nuts beneath the pump frame are not permitted. In cases where leveling is necessary, or where there are gaps between the pump frame and base substrate, solid metal shims or spacers must be used and anchors must be in full tension for rigid anchoring.

Pump Series: PL; BJ; PN

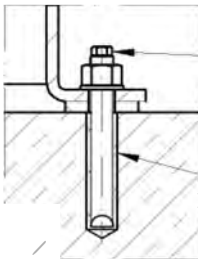


Anchor Bolts:
Hilti HAS-R 316 Stainless Steel
1/2" X 6-1/2"

Hilti HIT-HY 200 Adhesive

**Tags: 10-PMP-2210
10-PMP-2220**

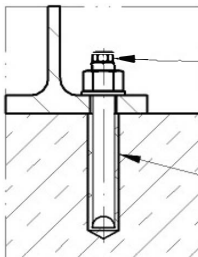
Pump Series: **CL**; FL; EL; BL; QN



Anchor Bolts:
Hilti HAS-R 316 Stainless Steel
5/8" X 9"

Hilti HIT-HY 200 Adhesive

Pump Series: XL



Anchor Bolts:
Hilti HAS-R 316 Stainless Steel
3/4" X 10"

Hilti HIT-HY 200 Adhesive

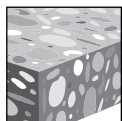
3.2.2 HIT-HY 200 ADHESIVE ANCHORING SYSTEM

PRODUCT DESCRIPTION

HIT-HY 200 with HIT-Z rods, Threaded Rod, Rebar, and HIS-N/RN Inserts

Anchor System	Features and Benefits
 Hilti HIT-HY 200-R Cartridge	- Two great products with equal performance data - User can select product gel time suitability based on temperature of the base material and jobsite time requirements - No hole cleaning requirement when installed SafeSet™ hollow drill bit technology
 Hilti HIT-HY 200-A Cartridge	- No hole cleaning requirement when installing HIT-Z anchor rods in dry conditions with hammer drilled holes - ICC-ES approved for cracked concrete and seismic service
 Hilti HIT-Z Anchor Rod	May be installed in diamond cored holes with HIT-Z anchor rod only when additional cleaning steps are employed
 Hilti HAS Threaded Rod	ICC-ES approved for grout-filled concrete masonry
 Rebar	
 Hilti HIS-N/RN	

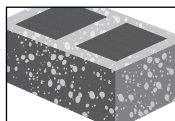
Supplied Anchor Rod:
HAS 5/8" x 9"



Uncracked concrete



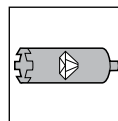
Cracked concrete



Grout-filled concrete masonry



Seismic Design Categories A-F



Diamond cored holes for Cracked and Uncracked Concrete

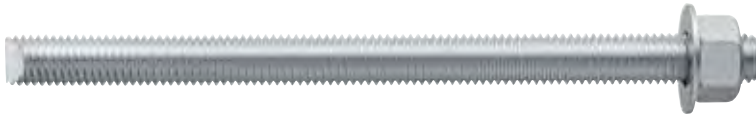


Hollow Drill Bit



Profis Anchor design software

Approvals/Listings	
ICC-ES (International Code Council)	ESR-3187 in concrete per ACI 318-14 Ch. 17 / ACI 355.2/ ICC-ES AC308 ESR-3963 in grout-filled CMU per ICC-ES AC58 ELC-3187 in concrete per CSA A23.3-14 / ACI 355.2
NSF/ANSI Std 61	Certification for use in potable water
European Technical Approval	ETA-11/0492, ETA-11/0493 ETA-12/0006, ETA-12/0028 ETA-12/0083, ETA-12/0084
City of Los Angeles	City of Los Angeles 2017 LABC Supplement (within ESR-3187 for Concrete) Research Report No. 26077 for Masonry
Florida Building Code	2017 Florida Building Code Supplement (within ESR-3187)
U.S. Green Building Council	LEED® Credit 4.1-Low Emitting Materials
Department of Transportation	Contact Hilti for various states

HIT-HY 200 Adhesive with HAS Threaded Rod

Hilti HAS threaded rod

Figure 9 - Hilti HAS threaded rod installation conditions

Permissible concrete conditions	Uncracked concrete	Dry concrete	Permissible drilling method	Hammer drilling with carbide tipped drill bit
	Cracked concrete	Water saturated concrete		Hilti TE-CD or TE-YD Hollow Drill Bit

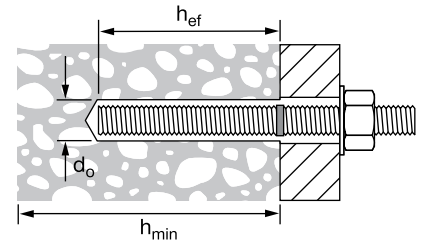
3.2.2

Table 38 - Hilti HAS threaded rod specifications

Setting information		Symbol	Units	Nominal rod diameter, d						
Nominal bit diameter		d_o	in.	7/16	9/16	5/8	3/4	7/8	1	1-1/4
Effective embedment	minimum	$h_{ef,min}$	in. (mm)	2-3/8 (60)	2-3/4 (70)	3-1/8 (79)	3-1/2 (89)	3-1/2 (89)	4 (102)	5 (127)
	maximum	$h_{ef,max}$	in. (mm)	7-1/2 (191)	10 (254)	12-1/2 (318)	15 (381)	17-1/2 (445)	20 (508)	25 (635)
Diameter of fixture hole	through-set		in.	1/2	5/8	13/16 ¹	15/16 ¹	1-1/8 ¹	1-1/4 ¹	1-1/2 ¹
Diameter of fixture hole	preset		in.	7/16	9/16	11/16	13/16	15/16	1-1/8	1-3/8
Installation torque		T_{inst}	ft-lb (Nm)	15 (20)	30 (40)	60 (80)	100 (136)	125 (169)	150 (203)	200 (271)
Minimum concrete thickness		h_{min}	in. (mm)	$h_{ef}+1-1/4$ ($h_{ef}+30$)		$h_{ef}+2d_o$				
Minimum edge distance		c_{min}	in. (mm)	1-3/4 (45)	1-3/4 (45)	2 ² (50) ²	2-1/8 ² (55) ²	2-1/4 ² (60) ²	2-3/4 ² (70) ²	3-1/8 ² (80) ²
Minimum anchor spacing		s_{min}	in. (mm)	1-7/8 (48)	2-1/2 (64)	3-1/8 (79)	3-3/4 (95)	4-3/4 (111)	5 (127)	6-1/4 (159)

¹ Install using (2) washers. See Figure 11.

² Edge distance of 1-3/4-inch (44mm) is permitted provided the installation torque is reduced to 0.30 T_{inst} for $5d < s < 16$ -in. and to 0.5 T_{inst} for $s > 16$ -in.

Figure 10 - Hilti HAS threaded rods**Figure 11 - Installation with (2) washers**

HIT-HY 200 Adhesive with Hilti HAS Threaded Rod

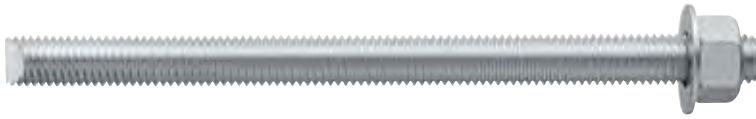


Table 81 - Steel factored resistance for Hilti HAS threaded rods for use with CSA A23.3-14 Annex D



Nominal anchor diameter in.	HAS-V-36 / HAS-V-36 HDG ASTM F1554 Gr.36 ^{4,6}			HAS-E-55 / HAS-E-55 HDG ASTM F1554 Gr. 55 ^{4,5,6}			HAS-B-105 / HAS-B-105 HDG ASTM A193 B7 and ASTM F 1554 Gr.105 ^{4,6}			HAS-R stainless steel ASTM F593 (3/8-in to 1-in) ⁵ ASTM A193 (1-1/8-in to 2-in) ⁴		
	Tensile ¹ ΦN_{sar} lb (kN)	Shear ² ΦV_{sar} lb (kN)	Seismic Shear ³ $\Phi V_{sar,eq}$ lb (kN)	Tensile ¹ ΦN_{sar} lb (kN)	Shear ² ΦV_{sar} lb (kN)	Seismic Shear ³ $\Phi V_{sar,eq}$ lb (kN)	Tensile ¹ ΦN_{sar} lb (kN)	Shear ² ΦV_{sar} lb (kN)	Seismic Shear ³ $\Phi V_{sar,eq}$ lb (kN)	Tensile ¹ ΦN_{sar} lb (kN)	Shear ² ΦV_{sar} lb (kN)	Seismic Shear ³ $\Phi V_{sar,eq}$ lb (kN)
3/8	3,055 (13.6)	1,720 (7.7)	1,030 (4.6)	3,955 (17.6)	2,225 (9.9)	1,560 (6.9)	6,570 (29.2)	3,695 (16.4)	2,585 (11.5)	4,610 (20.5)	2,570 (11.4)	1,800 (8.0)
1/2	5,595 (24.9)	3,150 (14.0)	1,890 (8.4)	7,240 (32.2)	4,070 (18.1)	2,850 (12.7)	12,035 (53.5)	6,765 (30.1)	4,735 (21.1)	8,445 (37.6)	4,705 (20.9)	3,295 (14.7)
5/8	8,915 (39.7)	5,015 (22.3)	3,010 (13.4)	11,525 (51.3)	6,485 (28.8)	4,540 (20.2)	19,160 (85.2)	10,780 (48.0)	7,545 (33.6)	13,445 (59.8)	7,490 (33.3)	5,245 (23.3)
3/4	13,190 (58.7)	7,420 (33.0)	4,450 (19.8)	17,060 (75.9)	9,600 (42.7)	6,720 (29.9)	28,365 (126.2)	15,955 (71.0)	11,170 (49.7)	16,920 (75.3)	9,425 (41.9)	6,600 (29.4)
7/8	18,210 (81.0)	10,245 (45.6)	6,145 (27.3)	23,550 (104.8)	13,245 (58.9)	9,270 (41.2)	39,150 (174.1)	22,020 (97.9)	15,415 (68.6)	23,350 (103.9)	13,010 (57.9)	9,105 (40.5)
1	23,890 (106.3)	13,440 (59.8)	8,065 (35.9)	30,890 (137.4)	17,380 (77.3)	12,165 (54.1)	51,360 (228.5)	28,890 (128.5)	20,225 (90.0)	30,635 (136.3)	17,065 (75.9)	11,945 (53.1)
1-1/4	38,225 (170.0)	21,500 (95.6)	12,900 (57.4)	49,425 (219.9)	27,800 (123.7)	19,460 (86.6)	82,175 (365.5)	46,220 (205.6)	32,355 (143.9)	37,565 (167.1)	21,130 (94.0)	12,680 (56.4)

1 Tensile = $A_{se,N} \phi f_{uts} R$ as noted in CSA A23.3-14 Eq. D.2.

2 Shear = $A_{se,V} \phi 0.60 f_{uts} R$ as noted in CSA A23.3-14 Eq. D.31.

3 Seismic Shear = $\alpha_{seis} V_{sar}$: Reduction factor for seismic shear only. See CSA A23.3 Annex D for additional information on seismic applications. Seismic shear for HIT-RE 500 V3

4 HAS-V, HAS-E (3/8-in to 1-1/4-in), HAS-B, and HAS-R (Class 1; 1-1/4-in) threaded rods are considered ductile steel elements (including HDG rods).

5 HAS-R (CW1 and CW2; 3/8-in to 1-in) threaded rods are considered brittle steel elements.

6 6 3/8-inch dia. threaded rods are not included in the ASTM F1554 standard. Hilti 3/8-inch dia. HAS-V, HAS-E, and HAS-E-B (incl. HDG) threaded rods meet the chemical composition and mechanical property requirements of ASTM F1554.

3.2.2

INSTALLATION INSTRUCTIONS

Installation Instructions For Use (IFU) are included with each product package. They can also be viewed or downloaded online at www.hilti.com. Because of the possibility of changes, always verify that downloaded IFU are current when used. Proper installation is critical to achieve full performance. Training is available on request. Contact Hilti Technical Services for applications and conditions not addressed in the IFU.

MATERIAL SPECIFICATIONS

Figure 15 - Hilti HIT-HY 200 adhesive cure time and working time (approx.)

HIT-HY 200-A					
		HAS HIS-N Rebar		HIT-Z ¹	
[°C]	[°F]	t _{work}	t _{cure}	t _{work}	t _{cure}
-10...-5	14...23	1.5 h	7 h	–	–
-4...0	24...32	50 min	4 h	–	–
1...5	33...41	25 min	2 h	–	–
6...10	42...50	15 min	1.25 h	15 min	1.25 h
11...20	51...68	7 min	45 min	7 min	45 min
21...30	69...86	4 min	30 min	4 min	30 min
31...40	87...104	3 min	30 min	3 min	30 min

HIT-HY 200-R					
		HAS HIS-N Rebar		HIT-Z ¹	
[°C]	[°F]	t _{work}	t _{cure}	t _{work}	t _{cure}
-10...-5	14...23	3 h	20 h	–	–
-4...0	24...32	2 h	8 h	–	–
1...5	33...41	1 h	4 h	–	–
6...10	42...50	40 min	2.5 h	40 min	2.5 h
11...20	51...68	15 min	1.5 h	15 min	1.5 h
21...30	69...86	9 min	1 h	9 min	1 h
31...40	87...104	6 min	1 h	6 min	1 h

¹ It is permitted to install Hilti HIT-HY 200 with HIT-Z anchor rod down to 14° F (-10° C) provided the drilled hole has the drilling dust fully removed. This can be done with Hilti TE-CD or TE-YD hollow drill bit or with cleaning procedures used with standard threaded rod.

Resistance of cured Hilti HIT-HY 200 to chemicals

Chemical		Behavior
Acetic acid	10%	+
Acetone		●
Ammonia	5%	+
Benzyl alcohol		–
Hydrochloric acid	10%	●
Chlorinated lime	10%	+
Citric acid	10%	+
Concrete plasticizer		+
De-icing salt (Calcium chloride)		+
Demineralized water		+
Diesel fuel		+
Drilling dust suspension pH 13.2		+
Ethanol	96%	–
Ethylacetate		–
Formic acid	10%	+
Formwork oil		+
Gasoline		+
Glycole		●
Hydrogen peroxide	10%	●
Lactic acid	10%	+
Machinery oil		+
Methylethylketon		●
Nitric acid	10%	●
Phosphoric acid	10%	+
Potassium Hydroxide pH 13.2		+
Sea water		+
Sewage sludge		+
Sodium carbonate 10%	10%	+
Sodium hypochlorite 2%	2%	+
Sulphuric acid	10%	+
	30%	+
Toluene		●
Xylene		●

Key: – non-resistant
+ resistant
● limited resistance

Samples of the HIT-HY 200 adhesive were immersed in the various chemical compounds for up to one year. At the end of the test period, the samples were analyzed. Any samples showing no visible damage and having less than a 25% reduction in bending (flexural) strength were classified as “Resistant.” Samples that had slight damage, such as small cracks, chips, etc. or reduction in bending strength of 25% or more were classified as “Limited Resistance” (i.e. exposed for 48 hours or less until chemical is cleaned up). Samples that were heavily damaged or destroyed were classified as “Non-Resistant.”

Note: In actual use, the majority of the adhesive is encased in the base material, leaving very little surface area exposed.

ORDERING INFORMATION

HIT-Z anchor rod

Description	Bit dia. (in.)	Min. embed. (in.)	Qty
HIT-Z 3/8 x 3-3/8	7/16	2-3/8	40
HIT-Z 3/8 x 4 3/8	7/16	2-3/8	40
HIT-Z 3/8 x 5 1/8	7/16	2-3/8	40
HIT-Z 3/8 x 6 3/8	7/16	2-3/8	40
HIT-Z 1/2 x 4 1/2	9/16	2-3/4	20
HIT-Z 1/2 x 6 1/2	9/16	2-3/4	20
HIT-Z 1/2 x 8	9/16	2-3/4	20
HIT-Z 5/8 x 6	3/4	3-3/4	12
HIT-Z 5/8 x 8	3/4	3-3/4	12
HIT-Z 5/8 x 9 1/2	3/4	3-3/4	12
HIT-Z 3/4 x 6-1/2	7/8	4	6
HIT-Z 3/4 x 8 1/2	7/8	4	6
HIT-Z 3/4 x 9 3/4	7/8	4	6



HIT-HY 200-A



HIT-HY 200-R

HIT-HY 200-A (accelerated working time)

Description	Package contents	Qty
HIT-HY 200-A (11.1 fl oz/330 ml)	Includes (1) foil pack with (1) mixer and 3/8 filler tube per pack	1
HIT-HY 200-A Master Carton (11.1 fl oz/330 ml)	Includes (1) master carton containing (25) foil packs with (1) mixer and 3/8 filler tube per pack	25
HIT-HY 200-A Combo (11.1 fl oz/330 ml)	Includes (1) master carton containing (25) foil packs with (1) mixer and 3/8 filler tube per pack and (1) HDM 500 Manual Dispenser	25
HIT-HY 200-A Master Carton (16.9 fl oz/500 ml)	Includes (1) master carton containing (20) foil packs with (1) mixer and 3/8 filler tube per pack	20
HIT-HY 200-A Combo (16.9 fl oz/500 ml)	Includes (2) master cartons containing (20) foil packs each with (1) mixer and 3/8 filler tube per pack and (1) HDM 500 Manual Dispenser	40
HIT-RE-M Static Mixer	For use with HIT-HY 200-A cartridges	1

HIT-HY 200-R (regular working time)

Description	Package contents	Qty
HIT-HY 200-R (11.1 fl oz/330 ml)	Includes (1) foil pack with (1) mixer and 3/8 filler tube per pack	1
HIT-HY 200-R Master Carton (11.1 fl oz/330 ml)	Includes (1) master carton containing (25) foil packs with (1) mixer and 3/8 filler tube per pack	25
HIT-HY 200-R Combo (11.1 fl oz/330 ml)	Includes (1) master carton containing (25) foil packs with (1) mixer and 3/8 filler tube per pack and (1) HDM 500 manual dispenser	25
HIT-HY 200-R Master Carton (16.9 fl oz/500 ml)	Includes (1) master carton containing (20) foil packs with (1) mixer and 3/8 filler tube per pack	20
HIT-HY 200-R Combo (16.9 fl oz/500 ml)	Includes (2) master cartons containing (20) foil packs each with (1) mixer and 3/8 filler tube per pack and (1) HDM 500 manual dispenser	40
HIT-RE-M Static Mixer	For use with HIT-HY 200-R cartridges	1

TE-CD Hollow Drill Bits

Order Description	Working length (in.)
Hollow Drill Bit TE-CD 1/2-13	8
Hollow Drill Bit TE-CD 9/16-14	9-1/2
Hollow Drill Bit TE-CD 5/8-14	9-1/2
Hollow Drill Bit TE-CD 3/4-14	9-1/2
Hollow Drill Bit TE-CD 16-A (Replacement collar)	

TE-YD Hollow Drill Bits

Order Description	Working Length (in.)
Hollow Drill Bit TE-YD 3/4-24	15-1/2
Hollow Drill Bit TE-YD 7/8-24	15-1/2
Hollow Drill Bit TE-YD 1-24	15-1/2
Hollow Drill Bit TE-YD 1 1/8-24	15-1/2
Hollow Drill Bit TE-YD 25-A (Replacement collar)	

For ordering information on anchor rods and inserts, dispensers, hole cleaning equipment and other accessories, see section 3.2.9.

Coatings

PU Single-Coat Paint HS 6044



Material description:

2-component high solid polyurethane single-coat paint with low VOC content (approx. 300 g/l).

Designated use:

Functional coating for interior and exterior use, primarily suited to the coating of iron, steel and cast iron without previous primer coating. Designed to achieve high-quality surfaces, good corrosion protection properties and excellent durability in the course of normal stress.

Properties:

- economical use of energy, material and operational hours
- outstanding stability (approx. 300 µm DFT)
- excellent protection against corrosion (C4 high)
- single-layer application without primer
- rapid drying time
- extremely weatherproof
- good resistance to chemicals

Binder system:

Aliphatic polyurethane resin

Pigment system:

Weatherproof coating pigments and special functional fillers

Solid content:

Approx. 79 wt. % in the finished mixture
(percentile is dependent on shade)

Flash point:

Approx. 25 °C (mixture)

Density:

Approx. 1,3 g / cm³ in the mixture
(percentile is dependent on shade)

Temperature resistance:

Up to approx. 120 °C (no long-term stress)

Pot life:

The prepared mixture can be applied for approx. 2 hrs under normal conditions.

Can size:

2,5-kg can incl. RICKERT PU Hardener HS 6040;
other can sizes by arrangement.

Degree of gloss:

Satin-gloss

Shades:

As per RAL, NCS, British Standard etc. and as per colour samples; also in the express service programme.

Coverage:

Approx. 3.5 m² / kg finished mixture with a resulting dry film thickness of around 150 µm.

Viscosity supplied:

Shear thinning

Storage:

Store in a cool and dry, yet frost-free, place

Mixing ratio:

6 parts by weight PU Single-Coat Paint HS 6044
1 part by weight PU Hardener HS 6040

Labelling and safety advice:

See safety data sheet

Thinning and cleaning agent:

RICKERT PU Thinner 0045

Surface preparation:

The surface must be properly prepared as well as being free from oil, grease and dirt. Mill scale, rust and old coatings should be removed completely by mechanical or chemical means. A result comparable to the standard grade of cleanliness SA 2½ (DIN EN ISO 12944) should be aimed for.

Very smooth surfaces (cold-rolled sheet metal, turned steel surfaces, aluminium, and the like) should be abraded as far as possible and prepared using RICKERT Metal Primer 2091 Rapid.

Galvanised surfaces should be cleaned by steam blasting if need be or passivated with ammoniacal solution by using a corundum plastic pad.

Detailed information is to be found in Information Sheets 5 and 6 of the Federal Committee for Paint and Protecting Agents (BFS).

Application:

Combine both components in the correct mixing ratio and mix thoroughly.

a) Brush or roller:

Apply coating undiluted as far as possible.

b) High-pressure spraying:

Dilute coating with PU Thinner 0045 to approx. 25 - 30 s DIN 6 mm and apply with a 1.6 to 2.5-mm nozzle at 2 - 5 bar pressure.

c) Airless / Airmix method:

Dilute coating with PU Thinner 0045 to approx. 45 s DIN 6 mm. The pressure should be approx. 140 bar and a 0,011" to 0,013" nozzle should be used.

Application conditions:

Object and ambient temperature should be at least 8° C. The relative humidity should not exceed 85%. The optimum application temperature is between 20° and 30° C.

Drying:

Approximate values, determined at 20° C and approx. 150 µm dry coat; deviating values lead to changed drying properties:

dust-dry:	approx. ½ hr	T1
tack-free:	approx. 5½ hrs	T3
touch dry:	approx. 7 hrs	T4

At lower temperatures, a longer drying time should be allowed for.

Forced drying:

Approximate values, determined at 60° C object temperature and approx. 150 µm dry coat; deviating values lead to changed drying properties:

dust-dry:	approx. 15 min.	T1
tack-free:	approx. 20 min.	T3

Cleaning the equipment:

Clean with RICKERT PU Thinner 0045 immediately after use.

The purpose of this data sheet is to advise you. While all details conform to the latest state of the art, we can accept no liability for the results obtained during use, due to the variety of applications and the different substrates involved.

The publication of this data sheet invalidates all previous data sheets for this product.

Status: March 2018



Warranty Certificate

Warranty Certificate

This is to certify that:

BOERGER warrants all its products against defects in material and workmanship for two years or for the period agreed on. This warranty covers all costs associated with the correction of the defect, including but not limited to removal of defective parts, new parts, labor including travel and subsistence, and shipping for a period of two years from the date of substantial completion. Boerger will provide a timely response to correct the defect by providing in a timely fashion, temporary equipment as necessary to replace warranted items requiring repair or replacement, when warranted items are in use and are critical to the treatment process, as defined by Owner.

This warranty does not cover damage or defects due to normal wear and tear, misuse, alteration, neglect or accident, unsuitable lubricants, chemical, electrochemical or electrical influence or weather or nature influence. The products have to be operated according to O&M Manual.

No allowance will be made for repairs or alterations on site, unless with prior written consent or approval. In the event that the products or equipment are altered or repaired by others without prior written approval, all warranties are void.

BOERGER cannot be held liable to any event for loss of profits or revenue, interest, loss by reason of shutdown or non-operation or other delays, increased expense of operation of the equipment, loss of use of power systems, cost of purchase or replacement power, or claims of purchaser or Customers of purchaser for service interruptions, environmental damage or any special indirect, incidental or consequential damages arising out of the contract, or any breach thereof or any defect therein, or failure thereof, or malfunction of the apparatus furnished hereunder.

The BOERGER warranty applies only to the original user. The Company will accept no liability whatsoever for this equipment if resold to a third or subsequent parties.

Equipment included in an order to BOERGER manufactured by others is warranted by the warranty of the original manufacturer.